Deployment and QA guide

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Step by step, on **Windows, Linux and macOS**: install a release, configure it, start the server and the console, and verify every part with what you should see, then tick the QA checklist. The optional cluster part does the same for three servers acting as one. A PDF copy of this guide is DEPLOYMENT_AND_QA_GUIDE.pdf.

This guide is the one place for the deployment and QA steps; the details behind them each have one home, which the steps link: OPERATIONS_GUIDE.md (installing, services, monitoring, troubleshooting), CLUSTER_GUIDE.md (Drishti*net in depth), CONFIGURATION.md (every setting), CLI_REFERENCE.md (every command), DATA_LAKE_GUIDE.md (lakes and loads). Developers building from source start with QUICKSTART_DEVELOPERS.md.

Conventions. `<drishti>` is the release folder (a clone, or the unpacked release). `<host>` is the machine's **real address** on the network (`hostname -I` on Linux, `ipconfig getifaddr en0` on macOS, `ipconfig` on Windows), for example `192.168.2.77`: give people that address, never `localhost`. Each step says what to do **in the UI**, **from the command line** and, where a file decides it, **in YAML**. The pictures were taken on a server and console started exactly as this guide says, on the demo lake.

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1. What you deploy

Process	What it does	Port	Health	Runs on
server	one Java program, `drishti-server-2.9.11.g-exec.jar`: reads the lakes and sources, computes every number (VaR, exposure, BI), lays out every screen	**18480**	`/actuator/health`	Java 21
console	the web pages people open, a Python program, `drishti-console/run_drishti_web.py`: renders what the server sends	**17480**	`/healthz`, `/readyz`	Python 3.13 in its own environment

Both read from **one release folder** and write everything under **one data folder** (`<drishti>/data`, or `DRISHTI_DATA_DIR`): the lakes, users, caches, logs. The two can run on one machine or on two; several servers can form a cluster (section 10). The full picture: OPERATIONS_GUIDE.md › What you are running.

2. Prerequisites, per operating system

Need	Windows 10/11, Server 2019+	Linux (Ubuntu 22.04+, RHEL 9+)	macOS 13+ (Apple silicon or Intel)
Java 21, a JDK, 64-bit	Temurin 21 **ZIP** unpacked to `C:\jdk-21` (no installer needed), or the MSI	`sudo apt install openjdk-21-jdk` / `sudo dnf install java-21-openjdk-devel`	`brew install openjdk@21`, or the Temurin 21 `.tar.gz`
Python 3.13	python.org's installer for Windows, or its embeddable ZIP with pip	`uv python install 3.13`, or the distribution's `python3.13`	`brew install python@3.13`, or `uv python install 3.13`
The console's environment	`py -3.13 -m venv drishti-console\.venv` then pip (no other tool)	`uv venv` (recommended) or `python3.13 -m venv`	the same as Linux
Scripts	PowerShell 5.1 or 7: `tools\windows\*.ps1`	bash	bash or zsh
Memory	8 GB free for the demo lake (the server's heap 4 GB, the console about 1 GB)	the same	the same
Disk	5 GB free: the release 1 GB, the demo lake 3.4 GB	the same	the same
Ports	18480 and 17480 open on the machine's firewall for the people who use it	the same	the same

Windows without any `.exe` beyond Java and Python. Drishti needs nothing else: no Hadoop (`winutils.exe`), no Docker, no MinIO, no uv. Java comes from the Temurin ZIP (`C:\jdk-21\bin\java.exe` is Java's own), Python from python.org; the console's environment is made with Python's own `venv` and `pip` (through the company's mirror: `%APPDATA%\pip\pip.ini`); every Windows script is a PowerShell `.ps1` run with `powershell -ExecutionPolicy Bypass -File …`. Native libraries the server unpacks itself (DLLs loaded by `java.exe`) are not programs and are allowed. What works differently on Windows: OPERATIONS_GUIDE.md › Drishti on Windows.

Check the machine:

```
java -version                    # openjdk version "21.…"
python3.13 --version             # Python 3.13.…  (Windows: py -3.13 --version)
```

Sizing beyond the demo. A server's heap grows with the trades held a day (about 4 GB for 400,000, 8 GB for a million or more). The sizing planner (`tools/sizing_planner.py`, and **Admin → Capacity** in the console) turns your volumes into memory, CPU and disk: CLUSTER_GUIDE.md › Sizing, OPERATIONS_GUIDE.md › Admin → Capacity.

3. Get the release

A. Build it from a clone (needs the JDK only; the repository carries Maven):

```
git clone https://github.com/ajsinha/drishti.git && cd drishti
./mvnw -q clean package -DskipTests -pl :drishti-app -am           # Windows: .\mvnw.cmd -q clean pa
ckage -DskipTests -pl :drishti-app -am
ls kernel/drishti-server/target/*-exec.jar                                 # kernel/drishti-server/tar
get/drishti-server-2.9.11.g-exec.jar
```

B. Copy a release built elsewhere (Java bytecode is the same on every operating system). The release folder holds:

Path	What
`kernel/drishti-server/target/drishti-server-2.9.11.g-exec.jar` (or `drishti-server.jar` at the top)	the server, every library inside
`config/`	packs, lakes, modules, lenses, engines, tables, BI files: the site's configuration
`drishti-console/`	the console's code and `requirements.txt` (its `.venv` is made on the machine, never copied)
`tools/`	`drishti.py` (the command line), `windows/*.ps1`, `lake/`
`data/testdata/`	the committed test data the demo lake is built from (23 MB of `.jsonl.gz`)
`docs/`, `LICENSE`, `CHANGELOG.md`, `RELEASE_NOTES.md`, `THIRD-PARTY-NOTICES.md`	shown by the console's Help
`deploy/`	the systemd unit, the drain scripts, Docker files, Prometheus alerts and the Grafana dashboard

To make that folder as a ZIP from a built clone:

```
git archive --format=zip --prefix=drishti/ -o drishti-2.9.11.g.zip HEAD
zip -g drishti-2.9.11.g.zip kernel/drishti-server/target/drishti-server-2.9.11.g-exec.jar      # Wind
ows: Compress-Archive -Update
```

Unpack it anywhere (`C:\drishti`, `/opt/drishti`, `~/drishti`); every command below runs from that folder.

The console's environment, once per machine:

```
# Linux and macOS
uv venv --python 3.13 drishti-console/.venv
uv pip install --python drishti-console/.venv/bin/python -r drishti-console/requirements.txt
```

```
# Windows (no uv)
py -3.13 -m venv drishti-console\.venv
drishti-console\.venv\Scripts\python.exe -m pip install -r drishti-console\requirements.txt
```

You should see pip end with `Successfully installed …`, and `drishti-console/.venv/bin/python tools/drishti.py doctor` (Windows: `drishti-console\.venv\Scripts\python.exe tools\drishti.py doctor`) end with `doctor: all clear`.

4. Configure

Everything has a working default; a site changes only what differs. Settings come from, in order: the server's built-in defaults, `application.local.yaml` beside the jar, environment variables, then command-line arguments (OPERATIONS_GUIDE.md › Where settings come from; every setting: CONFIGURATION.md).

What	Where	Default
The data folder	`DRISHTI_DATA_DIR`	`<drishti>/data`
The lakes	one file per lake, `config/lakes/<lake>.yaml` (and `config/lakes/<module>/`), each with its root folder	`demo` at `<data>/delta/demo`; `risklake`, `worldlake`, `newslake`, `referencelake`, `capitalmarket`, `regulators` ship empty
The demo lake	built by `lake demo` (section 5); `DRISHTI_DELTA_ROOT` moves it	`<data>/delta/demo`
Packs	`DRISHTI_PACKS` (comma list)	the seven of `config/packs/default-packs.yaml`
Modules	`config/modules/<module>.yaml` (`enabled: true/false`), or `DRISHTI_MODULES_<MODULE>_ENABLED`	banking, capitalmarkets and economy on; the core is always on
Lenses	`config/lenses/<module>/<lens>.yaml`; the lenses each person may open: `DRISHTI_LENSES`	risk, trading, market (banking), world (economy)
Engines	`config/engines/<engine>.yaml`, **Admin → Engines**	VaR, exposure, FRTB-SA, SIMM, the MR cube on; the data inbox off
Your jars (JDBC drivers, connector plugins)	the lib folder, `DRISHTI_LIB_PATH` (one place; jars found in the older `external_lib/java` or `drishti.sources.plugin-dir` are moved into it once, listed in `lib/moved-<date-time>.bak`, and Health says so)	`<data>/lib`
Secrets	the environment only, never a file in `config/`	none: set them (below)

Modules (YAML; `POST /api/v1/admin/modules` writes the same file; a change takes effect at the next start):

```
# config/modules/banking.yaml
enabled: true
module: { id: banking, domain: banking, title: Banking and risk }
```

Lenses. A lens is a way of seeing the product (Drishti Risk, Trading, Market, World): its Home, menus, commands and packs. The instance stays the full product; each person opens a lens for their session from Home. `DRISHTI_LENSES=risk,world,market,trading` offers the four (ADMIN_GUIDE.md › Modules, functions and lenses).

Secrets

Set them in the environment of both processes (Linux: `/etc/drishti/drishti.env`, mode 600; Windows: the service account's environment or the scheduled task's script; macOS: the launchd plist's `EnvironmentVariables`). Make each with `openssl rand -hex 32` (Windows: `py -3.13 -c "import secrets; print(secrets.token_hex(32))"`).

Variable	Read by	What it is
`DRISHTI_TOKEN_SECRET`	server and console	signs the tokens of signed-in people; the same on both, at least 32 bytes
`DRISHTI_SESSION_SECRET`	console	signs the console's session cookie
`DRISHTI_SECURITY_ENABLED=true`, `DRISHTI_AUTH_ENABLED=true`	server; console	sign-in on (off as shipped, for a developer's machine)
`DRISHTI_SEED_ADMIN`, `DRISHTI_SEED_USERNAME`, `DRISHTI_SEED_PASSWORD`	server	the first administrator, made at the first start with no users
`DRISHTI_NET_SECRET`	every server of a cluster	the cluster key (section 10)

The security checklist before anyone else uses the machine: OPERATIONS_GUIDE.md › Security checklist.

One node or three. One server and one console serve a team and the demo. Choose a cluster when one machine's memory or CPU is not enough, when the service must survive a machine's loss, or for upgrades without downtime (section 10).

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5. Build the demo lake

The demo lake is sixty business days of production-shaped test data (5,000 trades a day from each of ENDUR, CALYPSO and MUREX, and everything the packs read around them), ending on today's business date. Stop the server first, then:

```
drishti-console/.venv/bin/python tools/drishti.py lake demo                # Windows: drishti-consol
e\.venv\Scripts\python.exe tools\drishti.py lake demo
```

You should see one line per table as it is written (a progress line every few days of a large one), then:

```
Demo lake ready: <drishti>/data/delta/demo: 86 tables, 60 business days 2026-08-13 to 2026-11-06 (the
test data's 2026-07-08 to 2026-09-30, moved to end on 2026-11-06), built in 864 s
Feeds: <drishti>/data/feeds/fixing.jsonl (the file connector's sample feed)
Connector files:
  trade-endur: generated by the server at its first start (reads <drishti>/data/delta/demo)
  …
```

It takes about 15 minutes and 3.4 GB. Then check that what the server will read matches the disk:

```
$ drishti-console/.venv/bin/python tools/drishti.py lake check
  note  demo/climate.climate-profile: no table yet at ./data/delta/demo/climate/climate-profile (a wr
iter creates it as the lake file says)
  …
lake check: 9 lakes, 0 errors
```

Every option (`--as-of`, `--only`, `--lake`): CLI_REFERENCE.md › The demo lakes.

6. Start the server and the console

Start and stop Drishti with the scripts in **`<drishti>/bin`**: each one comes as a `.sh` for Linux and macOS (bash; also Apple's bash 3.2), a `.ps1` for Windows PowerShell 5.1 and PowerShell 7, and a `.cmd` that runs the `.ps1` from `cmd` or a double-click (no change to the execution policy). The three take the same options; on Windows an option may also be written `-Name value` (`--env-file` is `-EnvFile`). This section is the one place for the full list of options; bin/README.md is the short version.

6.1 The start and stop scripts (`bin/`)

Script	What it does
`start-server`	finds Java 21 or newer, the newest server jar, sets the temporary folder, the memory settings and the logs under the data folder, starts the server (in the foreground, or in the background with `--background`), waits for `/actuator/health` and prints the address other machines open. With `--node` it starts a node of a cluster
`start-console`	finds the console's Python (makes `drishti-console/.venv` on first use), points the console at the server (or a cluster's entry nodes), waits for `/healthz` and prints the address people open
`stop-server`	stops the server (or one node, or every node) **by its pid file only**, never by name: a graceful stop first (SIGTERM; on Windows a graceful `taskkill`), forced after a timeout; removes the pid file
`stop-console`	the same for the console
`status`	every process this machine started with `bin/`: pid, running or a stale pid, port, whether its health URL answers, its LAN address
`start-all`	one server and one console on this machine, in the background (calls `start-server`, then `start-console`)
`stop-all`	the console, then the server (`--all-nodes`: every node too)
`start-cluster`, `stop-cluster`	a cluster on one machine from a cluster file (`--cluster FILE`, default `deploy/stack/stack.yaml`): every node, then the console, or only those named
`drishti-node`	one node's drain before a stop and its readiness: the stop hook of the service unit and the images

`start-server` options

Option	Default	Meaning
`--port N`	`DRISHTI_PORT`, else `18480`	the server's port
`--heap SIZE`	`2g` (`4g` with `--demo`)	the heap limit (`-Xmx`): about 4 GB per 400,000 trades a day
`--jar FILE`	the newest `kernel/drishti-server/target/drishti-server-*-exec.jar`, then `drishti-server*.jar` in `<drishti>`	the server jar
`--java-home DIR`	`DRISHTI_JAVA_HOME`, `JAVA_HOME`, `java` on the `PATH`; on macOS then `/usr/libexec/java_home -v 21+`	the JDK (21 or newer; refused when older)
`--packs LIST`	`DRISHTI_PACKS`, else the server's own (the demo's ten with `--demo`)	the packs to load

Option	Default	Meaning
`--data-dir DIR`	`DRISHTI_DATA_DIR`, else `<drishti>/data`	the data folder; the pid, log and temporary files go under it
`--env-file FILE`	none	`KEY=VALUE` lines put into the server's environment: the secrets stay off the command line, and their values are never printed
`--demo`	off	the offline demo on the lake of section 5 (the ten packs, samples off, offline, sign-in on, the four lenses, the data inbox on) with **development secrets and the `drishti-dev-admin` account: not for production**; a secret you set yourself wins. `-DemoLakes`, the old Windows name, is the same
`--engine NAME`	the server's own (Windows: `native`)	the Delta engine: `native`, `hadoop` (needs `winutils.exe` on Windows) or `auto`
`--lake-root DIR`	the lake files' own roots	the Delta lake root (`DRISHTI_DELTA_ROOT`)
`--iceberg`, `--iceberg-root DIR`	off, `data/iceberg`	serve the stores from Iceberg tables: copies `config/connectors.examples/iceberg/*.yaml` into `config/connectors`
`--java-opt FLAG`	none	one more JVM flag (repeatable)
`--server-arg ARG`	none	one more server argument such as `--drishti.x=y` (repeatable; on Linux and macOS also everything after `--`)
`--workdir DIR`	`<drishti>`	the server's working folder
`--background`	off (the foreground; Ctrl+C stops it)	detach; the server writes its own rotated log
`--wait SECONDS`	`300`	how long to wait for the health answer
`--stop`		the same as `stop-server` (kept for the old `start-server.ps1 -Stop`)
`--node NAME`	none (a single server)	**a cluster node**: its id; its pid, log and temporary folder are named after it, so several run on one machine
`--node-dir DIR`	`<data>/nodes/<node>/net`, made with a `node.yaml` when missing	the node's Drishti*net folder (`node.yaml`, `gossip.yaml`, `peers/`; section 10.4)
`--node-data DIR`	the data folder	the node's own data folder (its identity database, saved members, inbox). Two servers never share one: the script refuses a second node on the same folder. Several nodes on one machine: one each, e.g. `data/nodes/<node>/data`
`--seed HOST:PORT`	none	a member to join through (repeatable, or comma-separated); written to `<node-dir>/peers/` when not there yet
`--advertise IP`	this machine's real LAN address	the address the node advertises. Never `127.0.0.1`, `localhost` or `0.0.0.0`: the script refuses them

`start-console` options

Option	Default	Meaning
`--server URLS`	`DRISHTI_SERVER_URL`, else this machine's LAN address `:18480` (with a warning)	the server, or a cluster's **entry nodes**, comma-separated (`http://<ip>:18480,http://<ip2>:18480`): the console uses the first that answers and **learns the other members from the cluster**, failing over to them. `-Backend`, the old Windows name, is the same
`--port N`	`DRISHTI_CONSOLE_PORT`, else `17480`	the console's port
`--data-dir DIR`	`DRISHTI_DATA_DIR`, else `<drishti>/data`	the data folder
`--env-file FILE`	none	`KEY=VALUE` lines (the token and session secrets, the same as the servers')
`--demo`	off	the demo's console settings with **development secrets: not for production** (sign-in on, plain-http cookies)
`--python FILE`	`python3` (Windows: `python`)	the Python 3.11+ that makes `drishti-console/.venv` when `uv` is not installed
`--reinstall`	off	remake `drishti-console/.venv` (after `drishti-console/requirements.txt` changed)
`--background`, `--wait SECONDS`	off, `90`	detach; how long to wait for `/healthz`

`stop-server`, `stop-console`, `status`, `start-all`, `stop-all` options

Script	Options
`stop-server`	`--node NAME` (one node; without it the single server), `--all-nodes` (every server pid file of the data folder), `--data-dir DIR`, `--env-file FILE`, `--timeout SECONDS` (before the forced stop; `60`)
`stop-console`	`--data-dir DIR`, `--env-file FILE`, `--timeout SECONDS` (`30`)
`status`	`--data-dir DIR` (repeatable), `--env-file FILE`
`start-all`	`--demo`, `--env-file FILE`, `--data-dir DIR`, `--server-port N`, `--console-port N`, `--heap SIZE`
`stop-all`	`--data-dir DIR`, `--env-file FILE`, `--all-nodes`
`start-cluster`, `stop-cluster`	`--cluster FILE`, `--data-dir DIR`, then node names (`console` for the console)
`status --cluster FILE`	also shows the members each node sees and the console's servers

Every script takes `--help` (`-h`; on Windows also `-Help`).

The environment they read: `DRISHTI_JAVA_HOME`, `JAVA_HOME`, `DRISHTI_PORT`, `DRISHTI_CONSOLE_PORT`, `DRISHTI_DATA_DIR`, `DRISHTI_PACKS`, `DRISHTI_SERVER_URL` (the console's server), `DRISHTI_PYTHON` (the console's Python), and the secrets `DRISHTI_TOKEN_SECRET`, `DRISHTI_SESSION_SECRET`, `DRISHTI_NET_SECRET` (a cluster node). The scripts set for the processes: `DRISHTI_DATA_DIR`, `DRISHTI_TMP_DIR`, `DRISHTI_PORT`, and for a node `DRISHTI_NET_DIR` and `DRISHTI_ADVERTISE_HOST` (which the `node.yaml` they make names: `advertise-url: http://${DRISHTI_ADVERTISE_HOST}:${DRISHTI_PORT}`); for the console `DRISHTI_BACKEND_URLS` and `DRISHTI_BACKEND_URL`. Every other setting of CONFIGURATION.md goes in the environment or the `--env-file`.

Where the files go (`<data>` is the data folder):

File	Single server	Node `<node>`	Console
process id (first line; then `port=`, `url=`, `health=`, `data=`)	`<data>/run/server.pid`	`<data>/run/server-<node>.pid`	`<data>/run/console.pid`
log (`--background`; the server rotates it, 100 MB a file)	`<data>/logs/server.log`	`<data>/logs/server-<node>.log`	`<data>/logs/console.log`
standard output (the JVM's own lines, a crash)	`<data>/logs/server.out` (Windows `server.out.log`, `server.err.log`)	`<data>/logs/server-<node>.out`	in `console.log`
temporary folder	`<data>/tmp/server`	`<data>/tmp/server-<node>`	`<data>/tmp/console`
heap dump, GC log	`<data>/dumps/server`, `<data>/logs/gc-server-<pid>.log`	`<data>/dumps/<node>`, `<data>/logs/gc-server-<node>-<pid>.log`	

Exit codes: `0` up (or stopped, or not running); `1` failed to start (or `status`: one is not running or not answering); `2` a wrong option; `3` already running, the port is in use, or a node's data folder is taken.

6.2 One server and a console

The quickest start, on one machine, with the demo's settings and its development secrets (for a QA machine; anyone else uses an env-file, 6.4):

Linux and macOS

```
cd <drishti>
bin/start-all.sh --demo                      # the server, then the console, both in the background
```

Windows (`cmd`, PowerShell or a double-click on the `.cmd`):

```
cd <drishti>
bin\start-all.cmd --demo
```

The same in two terminals, in the foreground (Ctrl+C stops each):

```
bin/start-server.sh --demo                                  # terminal 1 (Windows: bin\start-server.
cmd --demo)
bin/start-console.sh --demo --server http://<host>:18480    # terminal 2 (Windows: bin\start-consol
e.cmd --demo --server http://<host>:18480)
```

You should see (the warning names the development secrets in use):

```
Drishti server: drishti-server-2.9.11.g-exec.jar on Java 21, port 18480, heap 4g, packs core,market-data,…, data <drishti>/data
warning: DEVELOPMENT secrets in use (NOT FOR PRODUCTION): DRISHTI_SEED_PASSWORD DRISHTI_TOKEN_SECRET DRISHTI_SESSION_SECRET. Put your own in an --env-file for anything shared.
server: started (pid 1974114); log <drishti>/data/logs/server.log; waiting up to 300 s for http://127.0.0.1:18480/actuator/health
server: up on http://192.168.2.77:18480
Drishti console on port 17480, given http://192.168.2.77:18480 (using http://192.168.2.77:18480 first), data <drishti>/data
console: started (pid 1975138); log <drishti>/data/logs/console.log; waiting up to 90 s for http://127.0.0.1:17480/healthz
console: up on http://192.168.2.77:17480
```

Open the console's address (the last line) in a browser and sign in as `drishti-dev-admin` (password `drishti-dev-admin123`). In the server's log, within a minute:

```
INFO  … DrishtiApplication : Starting DrishtiApplication v2.9.11.g using Java 21.0.12.1 …
INFO  … IdentityDatabase  : identity database ready: jdbc:sqlite:<drishti>/data/identity/drishti.db …
WARN  … UserService       : Created development admin 'drishti-dev-admin' with the default password. Change it before any shared use.
INFO  … DrishtiApplication : Started DrishtiApplication in 14.108 seconds (process running for 14.888)
```

The first time on macOS, if a script says *permission denied*, make the scripts executable once: `chmod +x bin/*.sh` (a copy that lost its permissions; a clone keeps them). **On Windows**, run the `.cmd` (it passes `-ExecutionPolicy Bypass`), or `powershell -ExecutionPolicy Bypass -File bin\start-server.ps1 …`.

6.3 In the background: status and stop

```
bin/start-server.sh --env-file /etc/drishti/drishti.env --heap 8g --background
bin/start-console.sh --env-file /etc/drishti/drishti.env --server http://<host>:18480 --background
bin/status.sh
bin/stop-console.sh
bin/stop-server.sh
```

```
bin\start-server.cmd --env-file C:\drishti-secrets\drishti.env --heap 8g --background
bin\start-console.cmd --env-file C:\drishti-secrets\drishti.env --server http://<host>:18480 --background
bin\status.cmd
bin\stop-console.cmd
bin\stop-server.cmd
```

You should see from `status`, then from the stops:

```
process                    pid  state      port   health     address
server                 1974114  running    18480  UP         http://192.168.2.77:18480
console                1975138  running    17480  UP         http://192.168.2.77:17480
console: stopped (pid 1975138)
server: stopped (pid 1974114)
```

A start of something already running says so and exits `3`; a stop of something not running says so and exits `0`; a pid file whose process is gone (after a reboot, a crash) is a **stale pid**: `status` shows it, and the next stop or start removes it and says so.

6.4 The secrets in an env-file

Put the secrets and settings in a file only the service account can read, and give it to every script with `--env-file`: the values never appear on the command line, in `ps`, or in the scripts' output.

```
# /etc/drishti/drishti.env (Windows: C:\drishti-secrets\drishti.env); chmod 600; never under config/ or in git
DRISHTI_TOKEN_SECRET=<32+ random bytes: openssl rand -hex 32>        # the same value on every server and the console
DRISHTI_SESSION_SECRET=<32+ random bytes>
DRISHTI_NET_SECRET=<32+ random bytes>                                  # a cluster only: the cluster key, the same on every node
DRISHTI_SECURITY_ENABLED=true
DRISHTI_AUTH_ENABLED=true
DRISHTI_SECURE_COOKIE=false                                            # false only for plain http; behind TLS leave it on
DRISHTI_PACKS=core,market-data,trading,counterparty-risk,market-risk,world-economy,news-and-markets,us-equities,gleif,funds
DRISHTI_DEMO_ENABLED=false
DRISHTI_VAR_SYSTEM_BASKETS_SEED=true
DRISHTI_OFFLINE=true
DRISHTI_LENSES=risk,world,market,trading
DRISHTI_DATALOAD_ENABLED=true
DRISHTI_DATALOAD_OWNER=true
DRISHTI_SERVER_URL=http://<host>:18480                                 # the console's server (instead of --server)
```

Blank lines and `#` comments are allowed, as is a leading `export`; one pair of quotes around a value is removed. On Windows make the file with `openssl` from Git for Windows, or `[Convert]::ToHexString((1..32 | % { Get-Random -Max 256 }))` in PowerShell 7.

6.5 Without the scripts

Under a service manager (section 7), or to see exactly what runs: the scripts start `java -Djava.io.tmpdir=<data>/tmp/server -Xmx… -XX:MaxDirectMemorySize=… -XX:TrimNativeHeapInterval=60000 -Djdk.nio.maxCachedBufferSize=262144 -XX:NativeMemoryTracking=summary -XX:+HeapDumpOnOutOfMemoryError … -jar <jar> --server.port=<port>` with the environment above, and `drishti-console/.venv/bin/python drishti-console/run_drishti_web.py --server.port=<port>` with `DRISHTI_BACKEND_URLS`. The first lines each script prints say which jar, Java, port, packs and data folder it used.

Port	Who calls it	Change with
18480 (server)	the console, `drishti.py`, scripts, other servers of a cluster	`--port` on `start-server` (`DRISHTI_PORT`)
17480 (console)	people's browsers	`--port` on `start-console` (`DRISHTI_CONSOLE_PORT`)
gossip (cluster only)	other servers	`gossip-port` in the node's `node.yaml` (section 10.2)

TLS. Put both behind the company's reverse proxy (or the servers' own TLS) before anyone else uses them, and leave `DRISHTI_SECURE_COOKIE` on: OPERATIONS_GUIDE.md › TLS and the reverse proxy.

A cluster (several servers acting as one) is started node by node with `start-server --node`: the worked examples, three nodes on one machine and on three machines, are in section 10.6.

6.6 When a script stops with an error

The script says	Do this
`port 18480 is in use by another process` (exit 3)	another copy runs: `bin/status.sh` (is it ours?), then `bin/stop-server.sh`; something else on the port: pick another with `--port`
`no Java or Drishti runs on Java 21 or newer; … says: …`	install Temurin JDK 21 and set `DRISHTI_JAVA_HOME` (or pass `--java-home`); on macOS `/usr/libexec/java_home -V` lists the JDKs
`no server jar`	build it (`./mvnw -q package -DskipTests -pl :drishti-app -am`) or pass `--jar`
`already running (pid …)` (exit 3)	it is: `status` shows it; stop it first to restart
`stale … removed (pid … is not running)`	nothing to do: the process had ended (a reboot, a crash); see its log for why
the console's `no server answers yet at …`	the server is not up yet (the console keeps trying), or `--server` names the wrong address: `curl http://<host>:18480/actuator/health`
`--advertise 127.0.0.1 is a loopback or wildcard address`	give the machine's real address; the default is the default route's interface. A machine with several interfaces (VPN, Docker): `--advertise <the LAN one>`
`a cluster node needs the cluster key DRISHTI_NET_SECRET`	put it (32+ bytes, the same on every node) in the `--env-file`
`… already uses the data folder …: give this node its own with --node-data`	two nodes on one machine: `--node-data <data>/nodes/<node>/data` for each
macOS: `permission denied`	`chmod +x bin/*.sh`; *cannot be opened because the developer cannot be verified* does not apply (these are text scripts), but a downloaded copy may carry quarantine: `xattr -dr com.apple.quarantine <drishti>`
Windows: `… cannot be loaded because running scripts is disabled on this system`	run the `.cmd`, or `powershell -ExecutionPolicy Bypass -File bin\….ps1 …`

6.7 Starting a white-labelled instance

One build, several products: a distribution (`config/distributions/drishti.<name>.yaml`, DISTRIBUTIONS.md) decides the name, theme, modules, lenses, Home and Help. Without one you get Drishti All, exactly as in 6.2. What a distribution leaves out does not exist in it: Drishti Finance has no VaR command, menu, Help page or AI tool; Drishti Risk has no economy command; market data is in both.

Drishti Risk, one machine. Linux and macOS:

```
cd <drishti>
bin/start-all.sh --demo --distribution risk
```

Windows (`cmd` or PowerShell):

```
cd <drishti>
bin\start-all.cmd --demo --distribution risk
```

You should see the same lines as in 6.2, then, at `http://<host>:17480`: **Drishti Risk** in the top bar on the navy dark theme, the menus Views, Blotter, Positions, Risk, Counterparties, Markets, BI, Admin, Help; Home's cards `VAR BANK`, `VAR DESK RATES`; Help titled **Drishti Risk Help**. Typing `ECON CAN <GO>` answers *DRS-4023 ECON is not in Drishti Risk*. `bin/status.sh` (Windows `bin\status.cmd`):

```
process                    pid  state      port   health     distribution     address
server                 2031144  running    18480  UP         risk             http://192.168.2.77:18
480
console                2031702  running    17480  UP         risk             http://192.168.2.77:17
480
```

Drishti Finance, one machine. The same with `--distribution finance` (Windows: `bin\start-all.cmd --demo --distribution finance`). **You should see Drishti Finance** on the light theme, the menus Views, Economies, Markets, Companies, BI, Admin, Help; no Risk menu; Home's cards `USEQ AAPL`, `FUND SPY`, `ECON CAN`, `NEWS rates`; Help titled **Drishti Finance Help** with no VaR, exposure, FRTB or SIMM guide; `VAR BANK <GO>` answers *DRS-4023 VAR is not in Drishti Finance*; `CRV <GO>` (market data) works.

Drishti Genomics (the sample). `bin/start-all.sh --demo --distribution genomics` (Windows: `bin\start-all.cmd --demo --distribution genomics`). **You should see Drishti Genomics** on the green theme with only the kernel's menus (Views, BI, Admin, Help) and no finance anywhere; Admin → Health says the `genomics` module was asked for and is *not installed*. **Install the sample module:** build it (`./mvnw -q package -pl modules/genomics -am`) and copy `modules/genomics/target/dmod/genomics-0.1.0.dmod` into the data folder's `modules/`. **You should see** it install itself within seconds (auto-deploy on a single node): the Research lens with VARIANT, GENE and COHORT, and the lollipop visual in Build → Elements gallery.

Drishti Risk and Drishti Finance over one cluster (the owner's stack). In the stack file, a `consoles:` list beside the nodes:

```
consoles:
  - { name: risk,    port: 17480, distribution: risk }
  - { name: finance, port: 17481, distribution: finance }
```

then `bin/start-cluster.sh --cluster deploy/stack/stack.yaml` (Windows: `bin\start-cluster.cmd --cluster deploy/stack\stack.yaml`). **You should see** each node start, then `console-risk: up on http://<host>:17480` and `console-finance: up on http://<host>:17481`; `bin/status.sh --cluster deploy/stack/stack.yaml` lists `console-risk … risk` and `console-finance … finance`. The nodes run Drishti All; each console shows its own distribution. Without a stack file, start a second console by hand: `bin/start-console.sh --server http://<host>:18480 --port 17481 --distribution finance --name finance --background` (Windows the same with `bin\start-console.cmd`), and stop it with `bin/stop-console.sh --name finance`.

Drishti All (the default). No argument, or `--distribution all` (Windows: `bin\start-all.cmd --demo --distribution all`). **You should see** exactly 6.2: **Drishti** on the terminal theme, every menu, Help titled **Help**, the lens switch offering Risk, World, Market and Trading; `bin/status.sh` shows `all` for both processes.

By setting instead of an argument. `DRISHTI_DISTRIBUTION=finance` in the environment or the env-file (both programs read it), or a path to a site's own file (`DRISHTI_DISTRIBUTION=/etc/drishti/acme.yaml`; Windows `C:\drishti\acme.yaml`). On the server it is the setting `drishti.distribution` (`--`

`drishti.distribution=finance` as a server argument); on the console the setting `distribution` (`--distribution=finance` as a console argument). A console given its own keeps it whatever the server runs.

From the IDEs. The run configurations in `run-configs/` (IntelliJ IDEA opens the repository root and lists them under *Drishti*):

Run configuration	Program	What it sets
`Drishti server (Risk)`	the server on 18480	environment `DRISHTI_DISTRIBUTION=risk`
`Drishti server (Finance)`	the server on 18480	environment `DRISHTI_DISTRIBUTION=finance`
`Drishti server (All)`	the server on 18480	environment `DRISHTI_DISTRIBUTION=all` (and today's `DRISHTI_PACKS`)
`Drishti console (Risk)`	the console on 17480	argument `--distribution=risk`, `--server.port=17480`; temp folder `data/tmp/console-risk`
`Drishti console (Finance)`	the console on 17481	argument `--distribution=finance`, `--server.port=17481`; temp folder `data/tmp/console-finance`
`Drishti console (All)`	the console on 17480	argument `--distribution=all`

The server configurations run in **IntelliJ IDEA** (module `drishti-app`, so banking is on the class path). The console configurations are Python configurations: run them from IntelliJ IDEA with the Python plugin, or from **PyCharm** opened on the repository root (they run `drishti-console/run_drishti_web.py` from the repository root with the project interpreter, the console's `.venv`). PyCharm opened on `drishti-console/` keeps its own `Drishti console`; to run a distribution there, add `--distribution=finance` to its parameters (or `DRISHTI_DISTRIBUTION=finance` to its environment).

Two consoles against one server from the IDEs: start `Drishti server (All)` (every module; a Risk or Finance server would limit both consoles to its own), then `Drishti console (Risk)` and `Drishti console (Finance)`. **You should see** Drishti Risk at `http://localhost:17480` and Drishti Finance at `http://127.0.0.1:17481`, each with its own top bar, theme and Help. Open the second by another host name (or in another browser profile): browsers share sign-in cookies across the ports of one host. The same works on Linux, macOS and Windows: the configurations use `$PROJECT_DIR$` paths only.

Verify. Admin → Health, the Distribution line: the console's and the server's distribution and each module *loaded*. Admin → White label: every distribution side by side, the running one first.

---

7. Run them as services

Linux (systemd). The shipped unit is `deploy/systemd/drishti-server.service` (the layout `/opt/drishti`, the user `drishti`, secrets from `/etc/drishti/drishti.env`, a drain before every stop); the console's unit, the layout and the environment file are in OPERATIONS_GUIDE.md › Running as services:

```
sudo cp deploy/systemd/drishti-server.service /etc/systemd/system/
sudo systemctl daemon-reload && sudo systemctl enable --now drishti-server drishti-console
systemctl status drishti-server --no-pager                    # active (running)
```

Windows (a scheduled task, no service wrapper `.exe`). One task per process, run at start-up as the service account, whether or not anyone is signed in. The task runs a small PowerShell file that sets the environment and starts the script in the foreground, so the task's state is the process's:

```
# C:\drishti\run-server.ps1
Set-Location C:\drishti
$env:DRISHTI_JAVA_HOME = 'C:\jdk-21'
$env:DRISHTI_TOKEN_SECRET = (Get-Content C:\drishti-secrets\token-secret.txt -Raw).Trim()   # a folde
r only the service account can read
$env:DRISHTI_SECURITY_ENABLED = 'true'; $env:DRISHTI_DEMO_ENABLED = 'false'; $env:DRISHTI_VAR_SYSTEM_
BASKETS_SEED = 'true'
& .\bin\start-server.ps1 --heap 4g
```

```
$action  = New-ScheduledTaskAction -Execute 'powershell.exe' -Argument '-NoProfile -ExecutionPolicy B
ypass -File C:\drishti\run-server.ps1'
$trigger = New-ScheduledTaskTrigger -AtStartup
$settings = New-ScheduledTaskSettingsSet -RestartCount 3 -RestartInterval (New-TimeSpan -Minutes 1) -
ExecutionTimeLimit ([TimeSpan]::Zero)
Register-ScheduledTask -TaskName 'Drishti server' -Action $action -Trigger $trigger -Settings $settin
gs -User 'DOMAIN\svc-drishti' -Password '<its password>'
Start-ScheduledTask -TaskName 'Drishti server'
```

Do the same for the console with `run-console.ps1` (`bin\start-console.ps1`). Before stopping the task, drain the server: `bin\drishti-node.ps1 drain -Wait 240` (CLUSTER_GUIDE.md › Drain before a stop). `powershell.exe` is part of Windows: nothing is installed.

macOS (launchd). One agent (or daemon, under `/Library/LaunchDaemons`) per process:

```


<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dt
d">
<plist version="1.0"><dict>
  <key>Label</key><string>com.example.drishti-server</string>
  <key>WorkingDirectory</key><string>/Users/you/drishti</string>
  <key>ProgramArguments</key><array>
    <string>/opt/homebrew/opt/openjdk@21/bin/java</string><string>-Djava.io.tmpdir=data/tmp/server</s
tring>
    <string>-Xmx4g</string><string>-jar</string><string>kernel/drishti-server/target/drishti-server-
2.9.11.g-exec.jar</string>
  </array>
  <key>EnvironmentVariables</key><dict>
    <key>DRISHTI_SECURITY_ENABLED</key><string>true</string>
    <key>DRISHTI_TOKEN_SECRET</key><string>…</string>
  </dict>
  <key>RunAtLoad</key><true/><key>KeepAlive</key><true/>
  <key>StandardOutPath</key><string>/Users/you/drishti/data/logs/server.out</string>
  <key>StandardErrorPath</key><string>/Users/you/drishti/data/logs/server.out</string>
</dict></plist>
```

```
mkdir -p data/logs data/tmp/server
launchctl load ~/Library/LaunchAgents/com.example.drishti-server.plist
launchctl list | grep drishti                                  # a process id in the first column
```

The console's plist runs `drishti-console/.venv/bin/python drishti-console/run_drishti_web.py` the same way. Docker and Kubernetes: OPERATIONS_GUIDE.md › Docker Compose.

---

8. Verify, step by step

Do these in order after every installation and upgrade. Each says what to do and what you should see; the QA checklist lists the same with a box to tick. For the command-line steps make a token once (step 8.3) and keep it in a file only you can read:

```
export DRISHTI_SERVER=http://<host>:18480
export DRISHTI_TOKEN_FILE=~/drishti-token                          # Windows: $env:DRISHTI_TOKEN_FILE
= "$HOME\drishti-token"
```

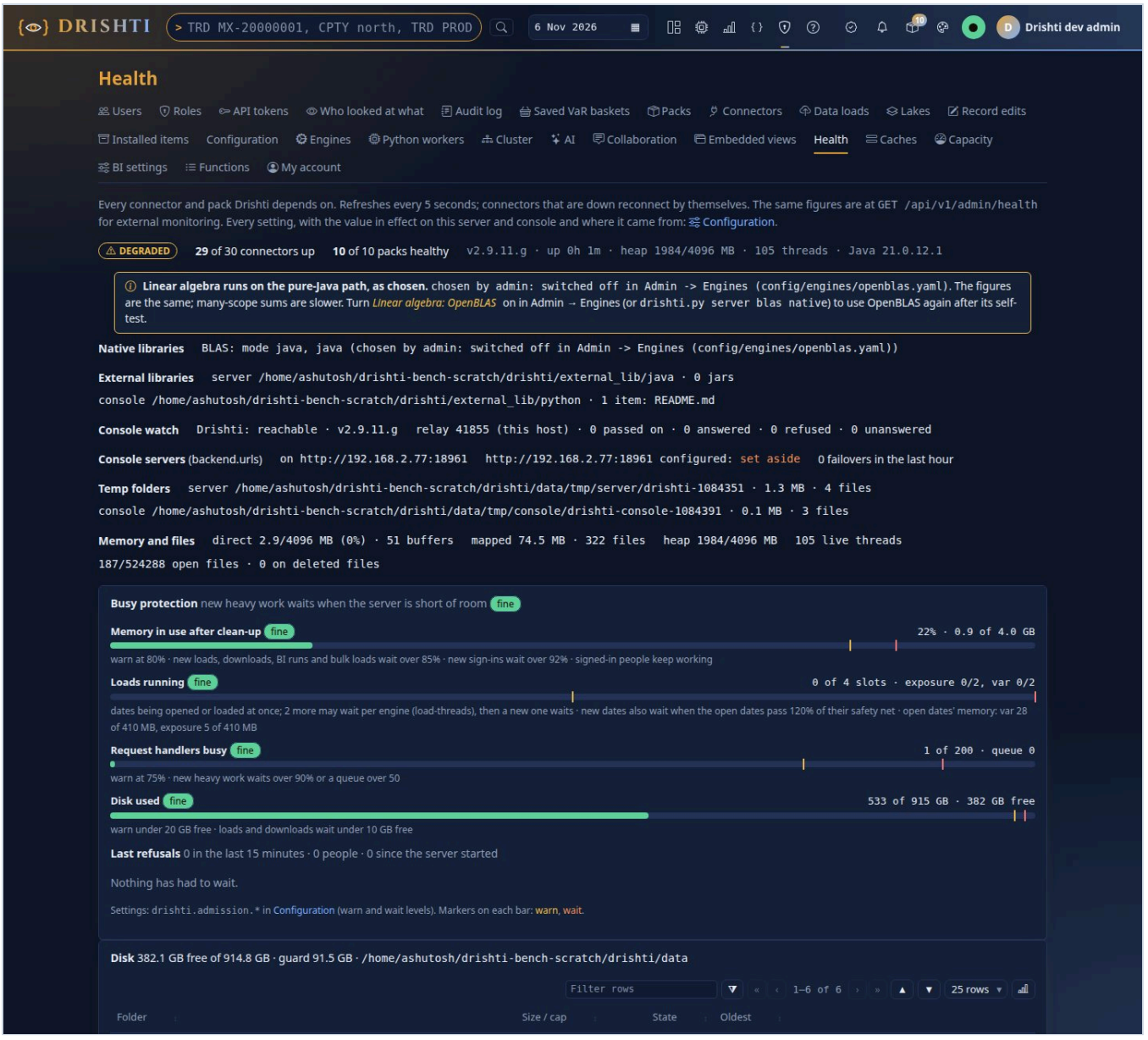
8.1 Health

From the command line: `bin/status.sh` (Windows `bin\status.cmd`) shows every process this machine started, its health answer and its address (exit 0 when all answer); each health address is open even with sign-in on:

```
$ bin/status.sh
process                    pid  state      port   health     address
server                 1974114  running    18480  UP         http://192.168.2.77:18480
console                1975138  running    17480  UP         http://192.168.2.77:17480
$ curl -s http://<host>:18480/actuator/health
{"status":"UP","groups":["liveness","readiness"]}
$ curl -s http://<host>:17480/readyz
{"status":"UP","server":"reachable"}
```

On Windows: `Invoke-RestMethod http://<host>:18480/actuator/health`.

In the UI, after signing in (8.3): **Admin → Health**: every line green or grey (not used), the server's version `2.9.11.g`, the lake connectors `UP`, the temporary folder under `data`.



8.2 The modules

From the command line (a token, 8.3):

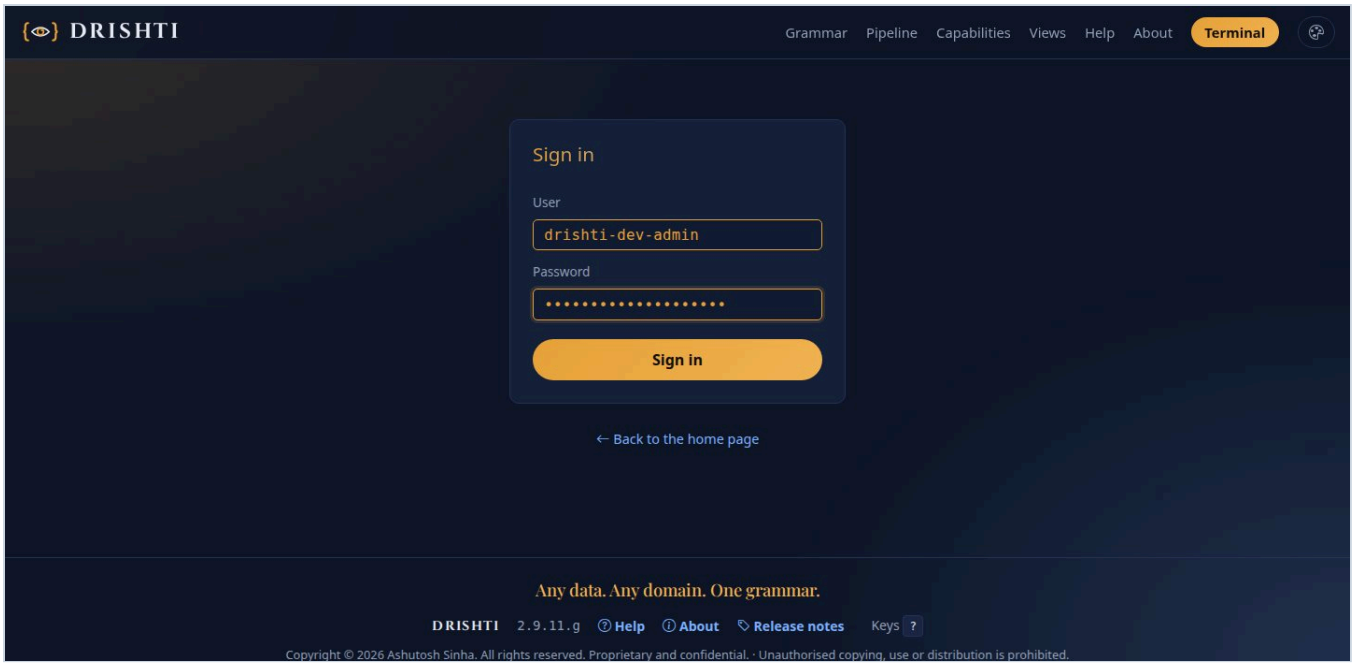
```
$ curl -s -H "Authorization: Bearer $(cat ~/drishti-token)" http://<host>:18480/api/v1/modules
{"modules":[{"id":"core","title":"Core","domain":"core"},
            {"id":"economy","title":"Economy and markets","domain":"economy"},
            {"id":"capitalmarkets","title":"Capital markets","domain":"capitalmarkets"},
            {"id":"banking","title":"Banking and risk","domain":"banking"}],
 "off":[],
 "lenses":[{"id":"world","title":"World","modules":["economy"],"functions":["ECON","MACRO",…],"hom
e":"home/world_landing.html",…},
           {"id":"publicsector",…}, {"id":"risk",…}, {"id":"trading",…}, {"id":"market",…}, …],
 "functions":{"off":[…]}}
```

You should see the modules core, economy, capitalmarkets and banking listed as loaded, `off` empty, and the lenses this person may open. `GET /api/v1/admin/modules` (administrators) adds each module's switch, licence and whether a restart is needed (OPERATIONS_GUIDE.md › Engines, modules and

editions).

8.3 Sign in, and a token for the command line

In the UI. Open `http://<host>:17480`: the sign-in page. The first start made the administrator of `DRISHTI_SEED_USERNAME` (`drishti-dev-admin`, password `drishti-dev-admin123` unless you set others): sign in, then change the password under your name → **My account** before anyone else uses the machine.



Then your name → **My account** → **API tokens** → a name, 30 days, no scope (read only) → **Create**; copy the secret (`drk_…`, shown once) into `~/drishti-token`.

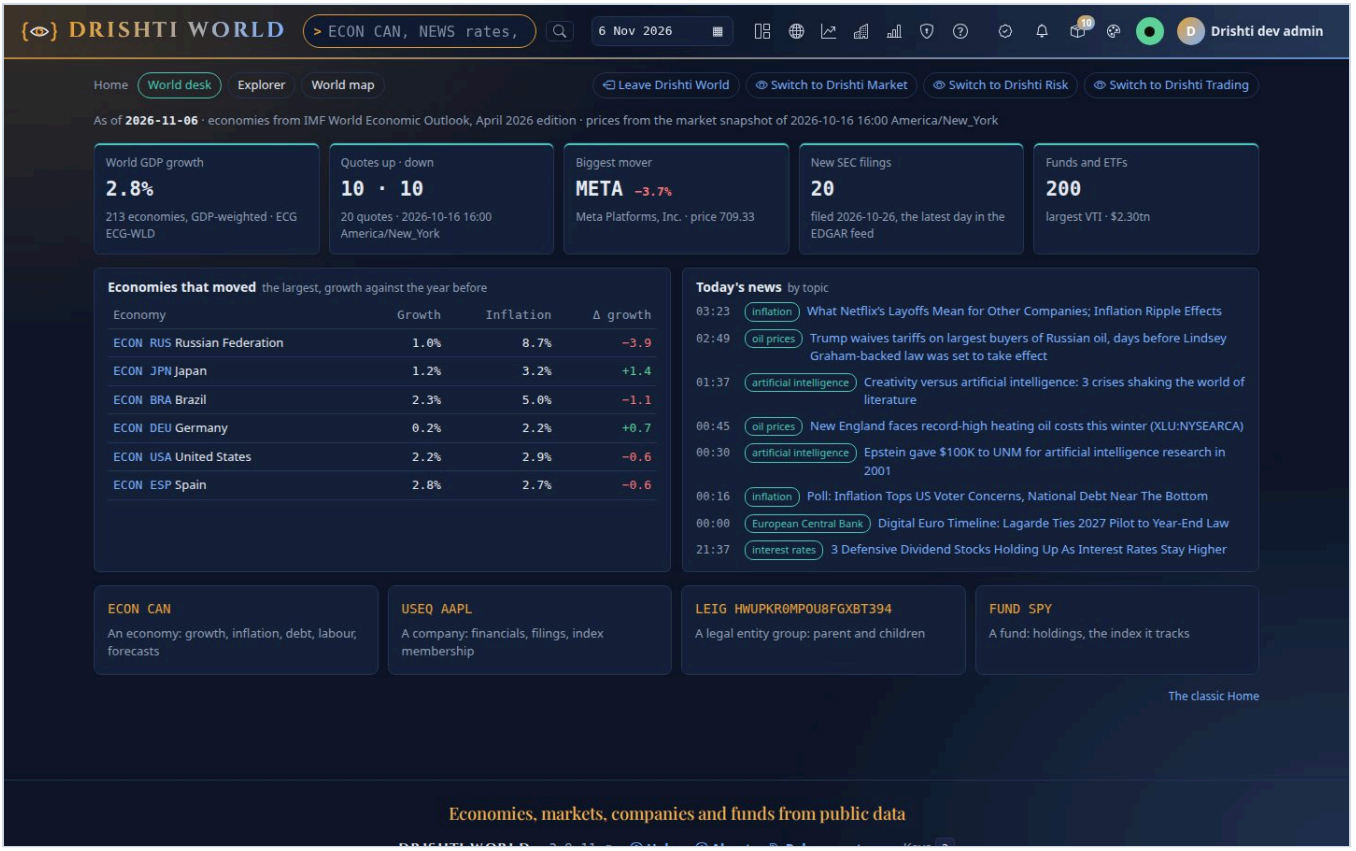
From the command line:

```
$ drishti-console/.venv/bin/python tools/drishti.py doctor --server http://<host>:18480 --token-file
~/drishti-token
GREEN  java                    …/bin/java: Java 21.0
GREEN  server jar              <drishti>/kernel/drishti-server/target/drishti-server-2.9.11.g-exec.ja
r (version 2.9.11.g)
GREEN  python                  <drishti>/drishti-console/.venv/bin/python: Python 3.13.15
…
GREEN  server                  http://<host>:18480: version 2.9.11.g, java 21.0.12.1, status UP
GREEN  token                   accepted; this token '…' acts as drishti-dev-admin: scopes read
doctor: 0 red, 0 yellow
```

8.4 The Home of each lens

In the UI. Home (the logo, top left) → **Open Drishti Risk**: the risk Home with the bank's VaR and exposure cards and the risk commands; then **Switch to Drishti World**: the world Home (economies, markets, news). **Leave Drishti World** returns to the full product.

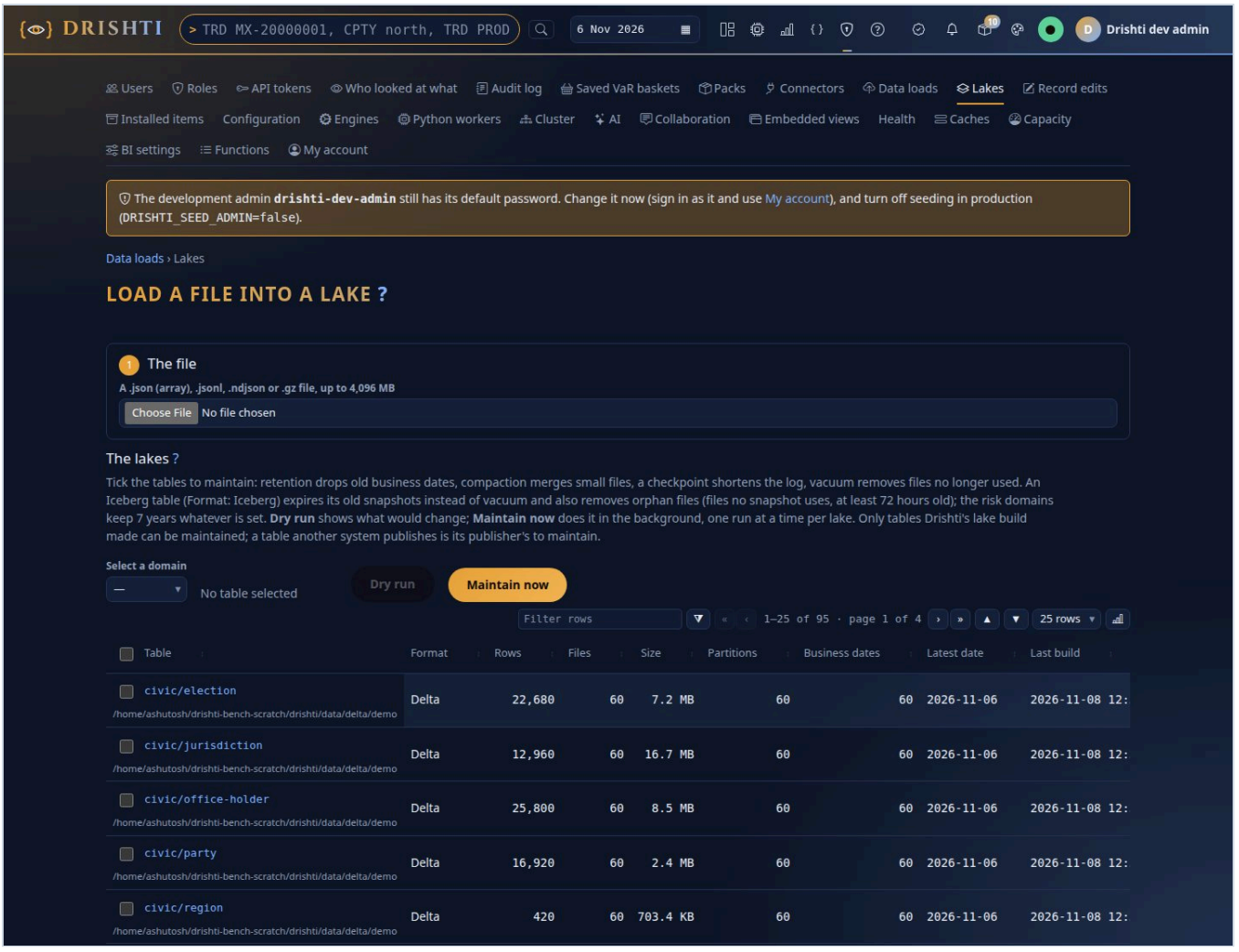




In YAML the Home of a lens is its file's `home:` block (`config/lenses/banking/risk.yaml`, `config/lenses/economy/world.yaml`).

8.5 The demo lake is loaded

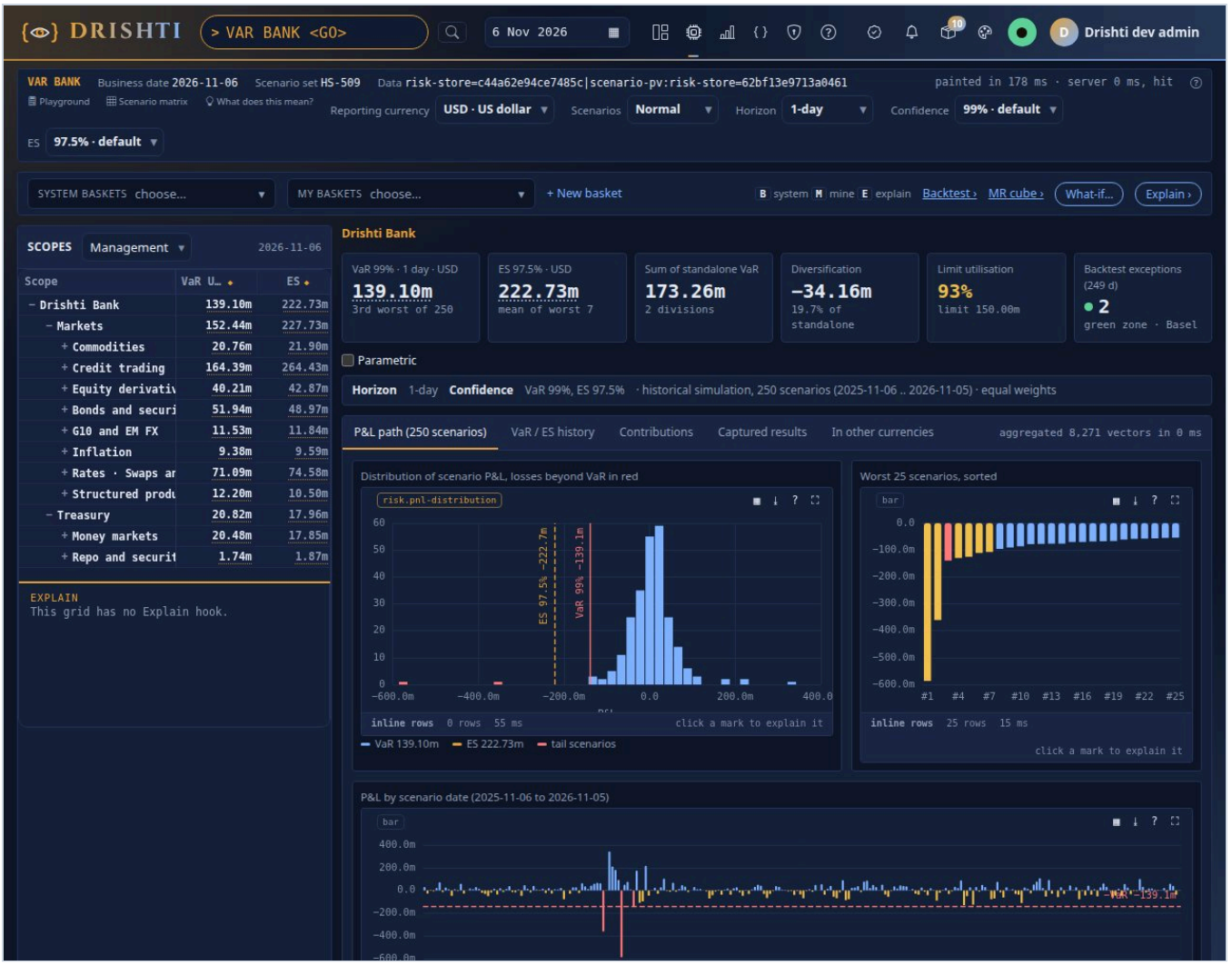
In the UI. Admin → Lakes: the `demo` lake with its tables (89), each with rows, files, size and its first and last business date; the other lakes listed empty.

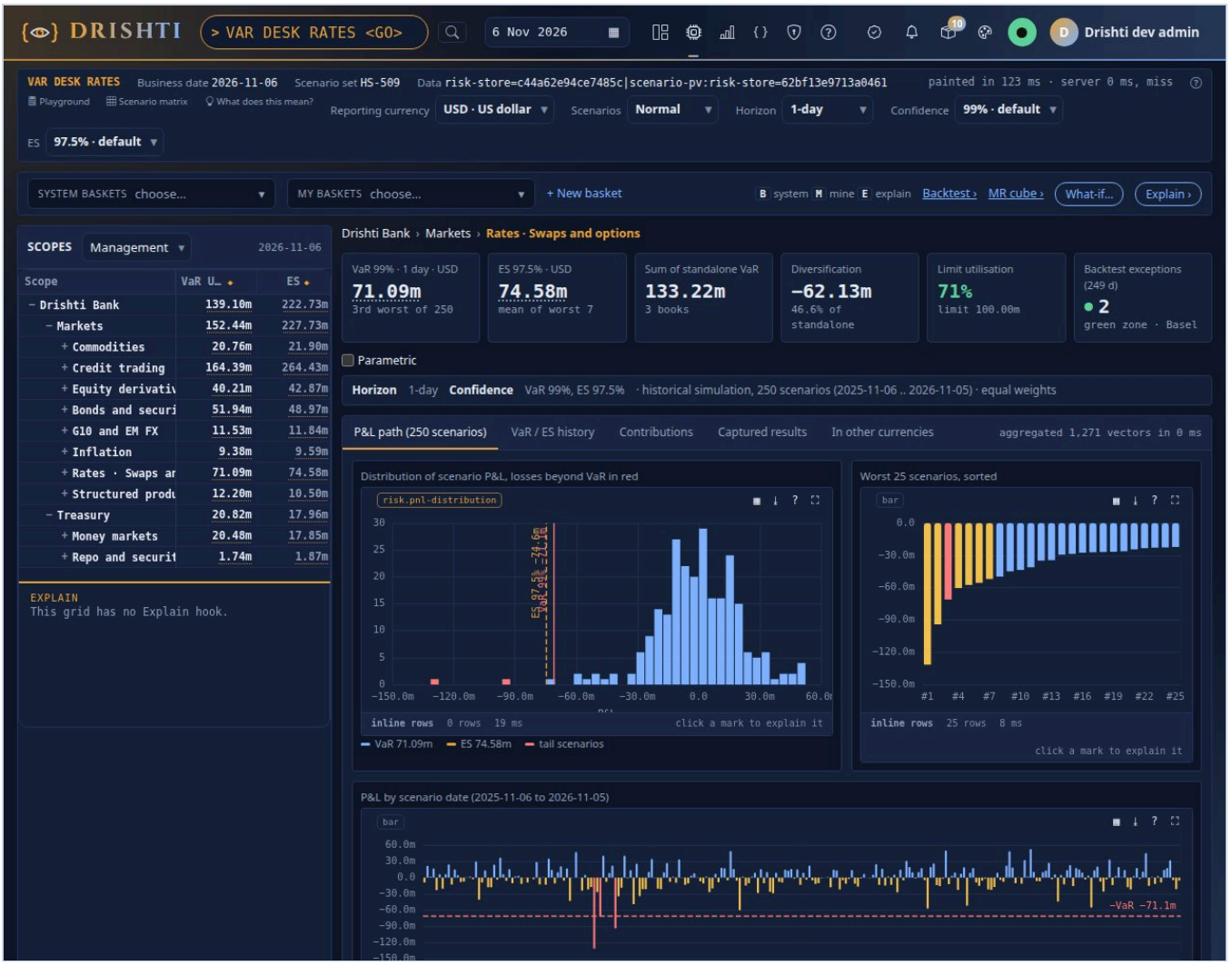


From the command line: `tools/drishti.py lake check` (section 5) prints the same lakes and says each connector names a known lake.

8.6 A VaR answer: the bank, then a desk

In the UI. Type `VAR BANK` in the command line (or **Risk → VaR explorer**) and press Enter: the bank's historical VaR and expected shortfall, the scope tree with every node's figure and limit bar, the P&L path. Then `VAR DESK RATES`: one desk, the tree below it with a breadcrumb up. **Explain** › on a row or on the VaR tile opens how the figure was built; **Why it moved** compares it with the previous day (or a week, a month end, a date you pick) in steps that add up (VAR_GUIDE.md section 3.2, section 3.11 (docs/guides/VAR_GUIDE.md#311-why-it-moved-the-var-change-explained)).


DRISHTI
> VAR BANK <GO>
6 Nov 2026
Drishti dev admin
VAR BANK
Business date 2026-11-06
Scenario set HS-509
Data risk-store=c44a62e94ce7485c|scenario-pv:risk-store=62bf13e9713a0461
painted in 178 ms · server 0 ms, hit
Playground
Scenario matrix
What does this mean?
Reporting currency
USD · US dollar
Scenarios
Normal
Horizon
1-day
Confidence
99% · default
ES
97.5% · default
SYSTEM BASKETS choose...
MY BASKETS choose...
+ New basket
B system M mine E explain
Backtest ›
MR cube ›
What-if...
Explain ›
SCOPES
Management
2026-11-06
Scope
VaR U…
ES
– Drishti Bank 139.10m 222.73m
– Markets 152.44m 227.73m
+ Commodities 20.76m 21.90m
+ Credit trading 164.39m 264.43m
+ Equity derivati 40.21m 42.87m
+ Bonds and securi 51.94m 48.97m
+ G10 and EM FX 11.53m 11.84m
+ Inflation 9.38m 9.59m
+ Rates · Swaps ar 71.09m 74.58m
+ Structured prod 12.20m 10.50m
– Treasury 20.82m 17.96m
+ Money markets 20.48m 17.85m
+ Repo and securit 1.74m 1.87m
EXPLAIN
This grid has no Explain hook.
Drishti Bank
VaR 99% · 1 day · USD
139.10m
3rd worst of 250
ES 97.5% · USD
222.73m
mean of worst 7
Sum of standalone VaR
173.26m
2 divisions
Diversification
−34.16m
19.7% of standalone
Limit utilisation
93%
limit 150.00m
Backtest exceptions (249 d)
2
green zone · Basel
Parametric
Horizon 1-day Confidence VaR 99%, ES 97.5% · historical simulation, 250 scenarios (2025-11-06 .. 2026-11-05) · equal weights
P&L path (250 scenarios)
VaR / ES history
Contributions
Captured results
In other currencies
aggregated 8,271 vectors in 0 ms
Distribution of scenario P&L, losses beyond VaR in red
risk.pnl-distribution
ES 97.5% −222.7m
VaR 99% −139.1m
inline rows 0 rows 55 ms
click a mark to explain it
VaR 139.10m
ES 222.73m
tail scenarios
Worst 25 scenarios, sorted
bar
inline rows 25 rows 15 ms
click a mark to explain it
P&L by scenario date (2025-11-06 to 2026-11-05)
bar
−VaR −139.1m



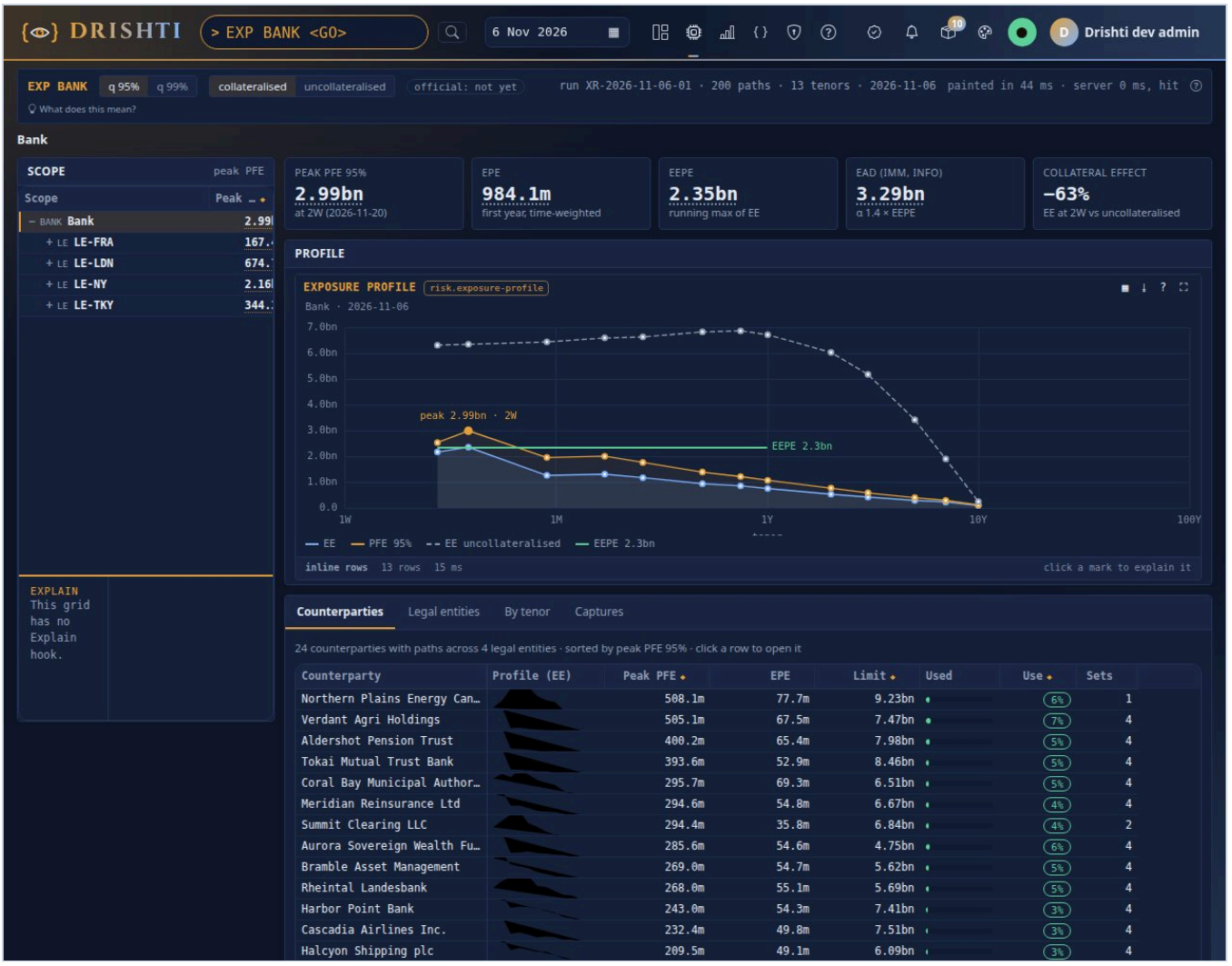
From the command line:

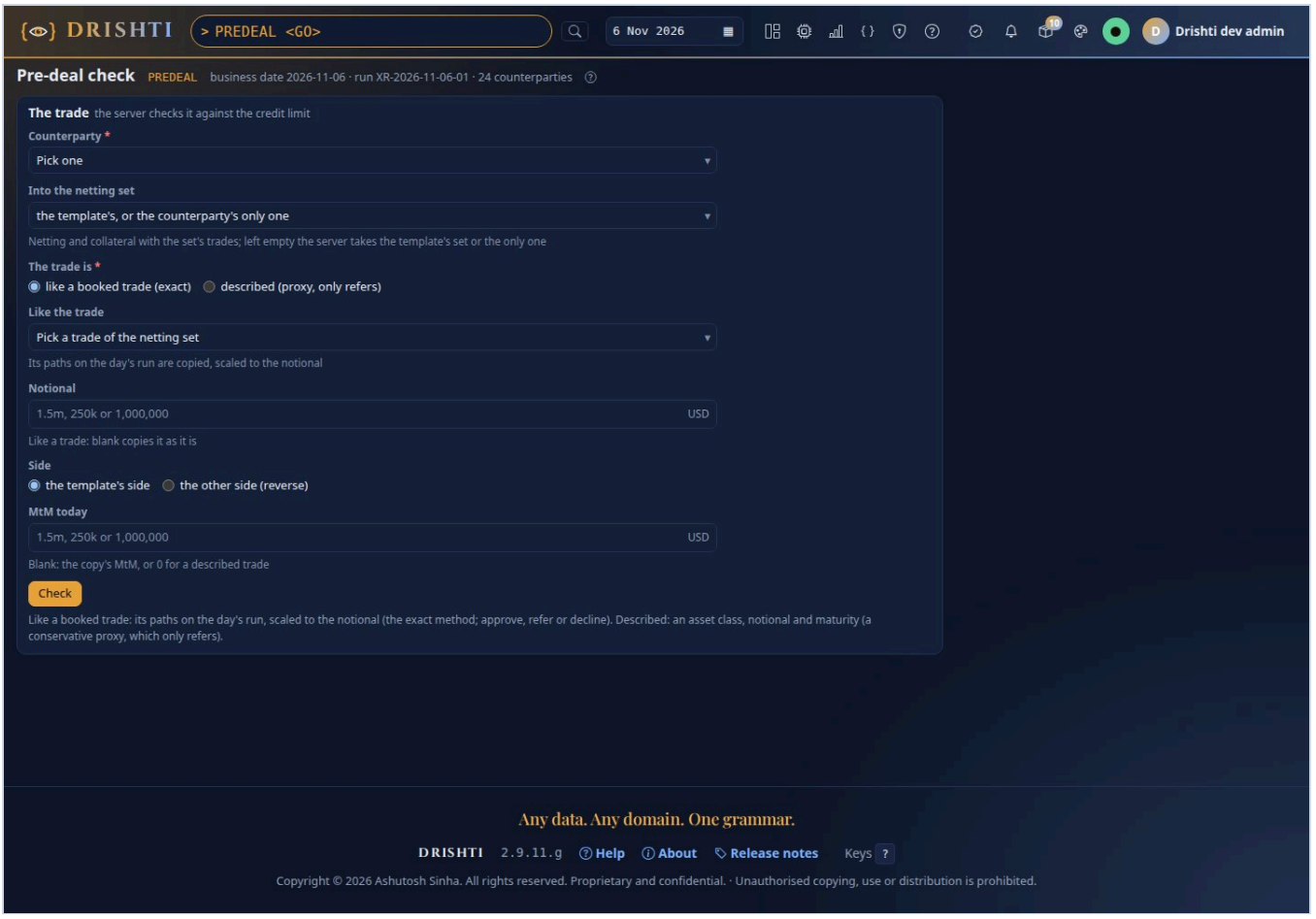
```
$ drishti-console/.venv/bin/python tools/drishti.py var show BANK
VAR BANK  ·  Drishti Bank  ·  node  ·  2026-11-06
  horizon    1-day   (1-day · 99% VaR, ES 97.5% · historical simulation, 250 scenarios)
  VaR 99%   139,096,039 USD   (the 3th worst of 250)
  ES 97.5%   222,732,223 USD   (mean of the 7 worst)
  limit      150,000,000 USD   use 92.7%
  parts      2 division   standalone sum 173,258,800   diversification 34,162,761
  backtest   green  2 exceptions in 249 days   Kupiec p 0.747   Christoffersen p 0.857
  8271 trades in 6876 positions
  …
```

The figures and how they are computed: VAR_GUIDE.md.

8.7 Exposure and the pre-deal check

In the UI. `EXP BANK`: the counterparty exposure explorer at the bank, its legal entities and counterparties with PFE and limits. Then **Counterparties → Pre-deal check** (or `PREDEAL`): a proposed trade checked against a counterparty's limit before it is dealt.

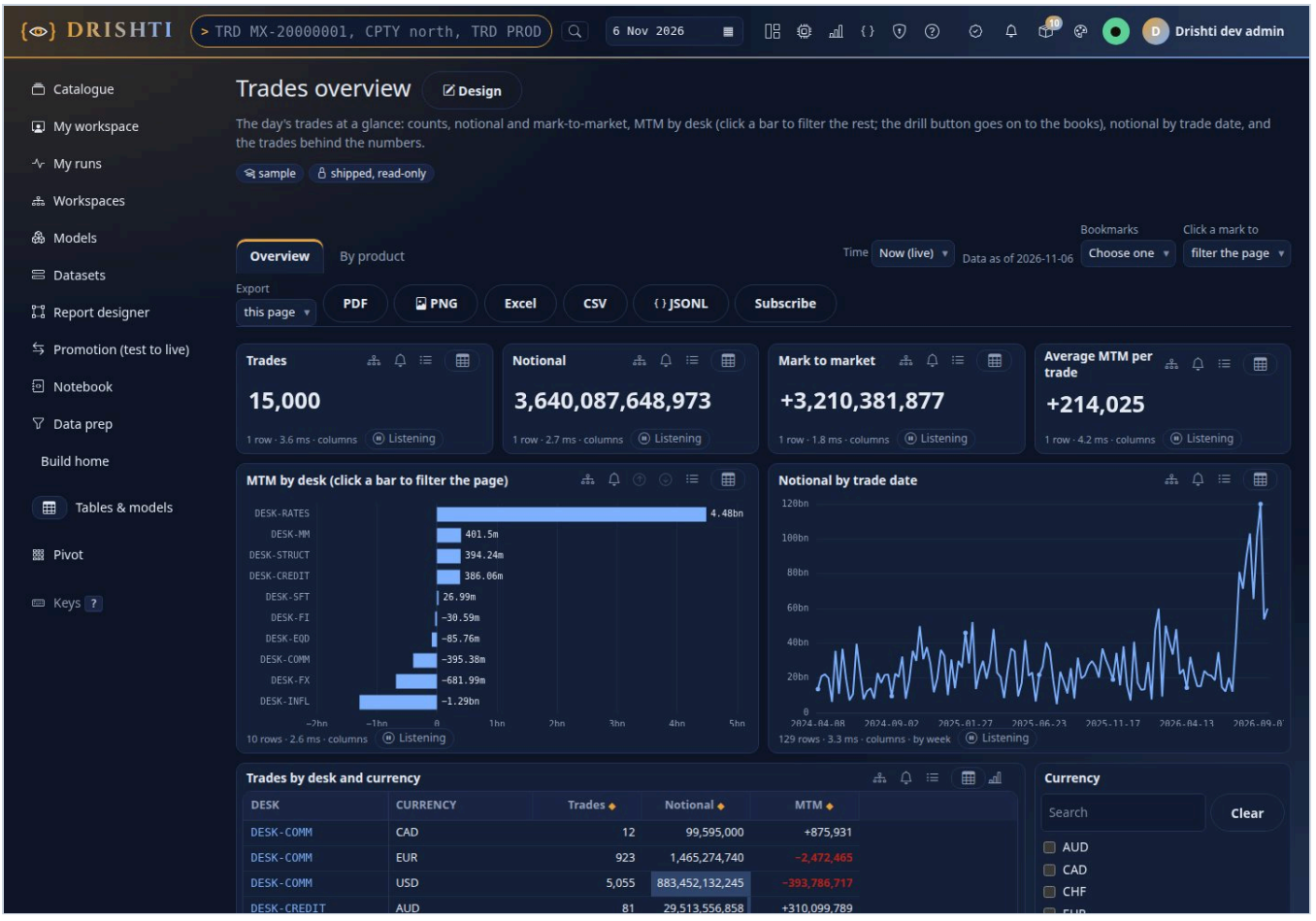

DRISHTI
> EXP BANK <GO>
6 Nov 2026
Drishti dev admin
EXP BANK
q 95%
q 99%
collateralised
uncollateralised
official: not yet
run XR-2026-11-06-01 · 200 paths · 13 tenors · 2026-11-06 painted in 44 ms · server 0 ms, hit
What does this mean?
Bank
SCOPE
peak PFE
Scope
Peak
BANK Bank
LE LE-FRA
LE LE-LDN
LE LE-NY
LE LE-TKY
PEAK PFE 95%
2.99bn
at 2W (2026-11-20)
EPE
984.1m
first year, time-weighted
EEPE
2.35bn
running max of EE
EAD (IMM, INFO)
3.29bn
α 1.4 × EEPE
COLLATERAL EFFECT
−63%
EE at 2W vs uncollateralised
PROFILE
EXPOSURE PROFILE
risk.exposure-profile
Bank · 2026-11-06
7.0bn
6.0bn
5.0bn
4.0bn
3.0bn
2.0bn
1.0bn
0.0
peak 2.99bn · 2W
EEPE 2.3bn
1W
1M
1Y
10Y
100Y
EE
PFE 95%
EE uncollateralised
EEPE 2.3bn
inline rows
13 rows
15 ms
click a mark to explain it
EXPLAIN
This grid has no Explain hook.
Counterparties
Legal entities
By tenor
Captures
24 counterparties with paths across 4 legal entities · sorted by peak PFE 95% · click a row to open it
Counterparty
Profile (EE)
Peak PFE
EPE
Limit
Used
Use
Sets
Northern Plains Energy Can… 508.1m 77.7m 9.23bn 6% 1
Verdant Agri Holdings 505.1m 67.5m 7.47bn 7% 4
Aldershot Pension Trust 400.2m 65.4m 7.98bn 5% 4
Tokai Mutual Trust Bank 393.6m 52.9m 8.46bn 5% 4
Coral Bay Municipal Author… 295.7m 69.3m 6.51bn 5% 4
Meridian Reinsurance Ltd 294.6m 54.8m 6.67bn 4% 4
Summit Clearing LLC 294.4m 35.8m 6.84bn 4% 2
Aurora Sovereign Wealth Fu… 285.6m 54.6m 4.75bn 6% 4
Bramble Asset Management 269.0m 54.7m 5.62bn 5% 4
Rheintal Landesbank 268.0m 55.1m 5.69bn 5% 4
Harbor Point Bank 243.0m 54.3m 7.41bn 3% 4
Cascadia Airlines Inc. 232.4m 49.8m 7.51bn 3% 4
Halcyon Shipping plc 209.5m 49.1m 6.09bn 3% 4

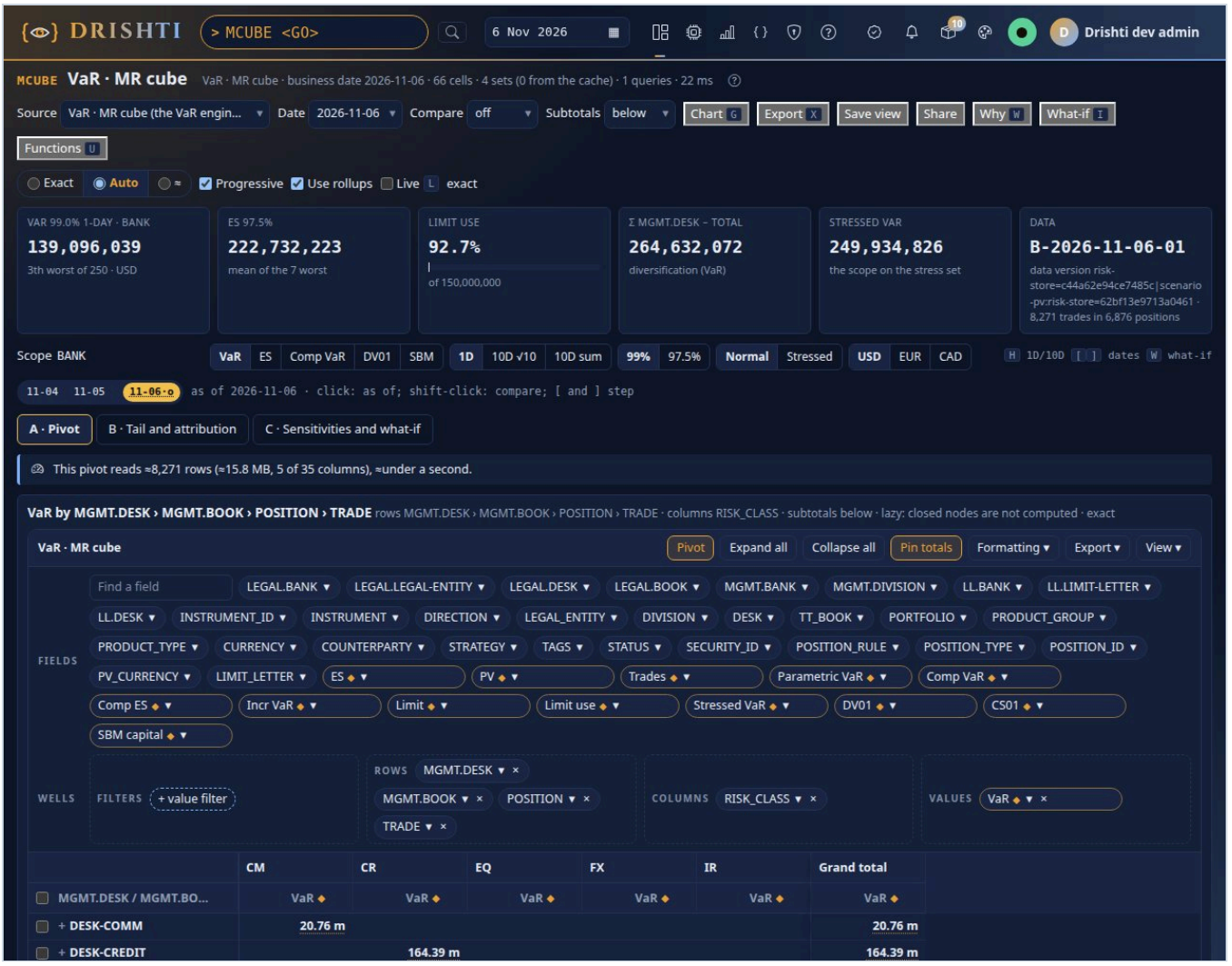


Details: EXPOSURE_GUIDE.md.

8.8 A BI report and the MR cube

In the UI. BI → Catalogue → Trades overview: the report's pages over the sixty days, its slicers live. Then `MCUBE`: the market-risk cube, the Pivot screen on the VaR source with VaR and ES by desk (a total is never the sum of its rows).

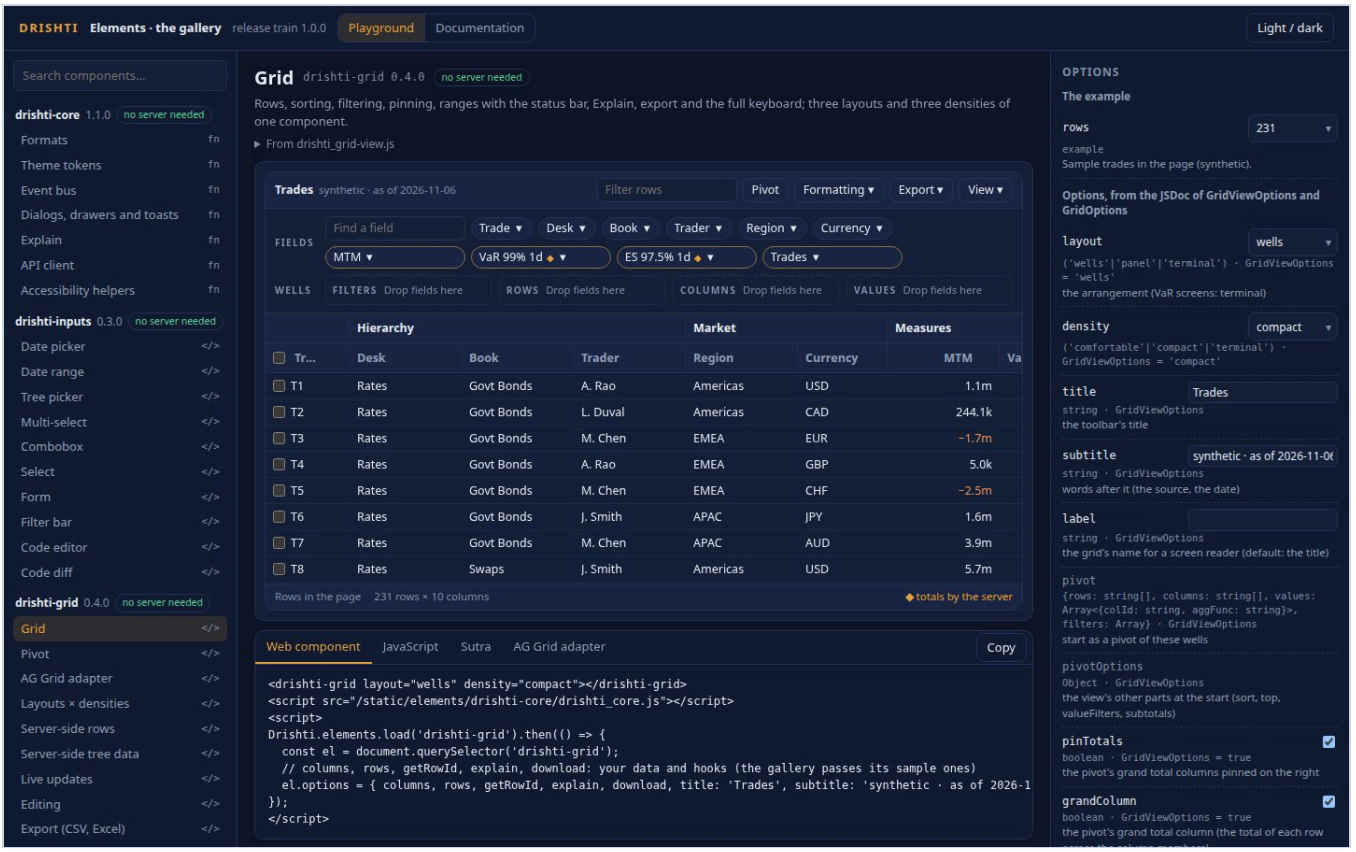

DRISHTI
> TRD MX-20000001, CPTY north, TRD PROD
6 Nov 2026
Drishti dev admin
Catalogue
My workspace
My runs
Workspaces
Models
Datasets
Report designer
Promotion (test to live)
Notebook
Data prep
Build home
Tables & models
Pivot
Keys
Trades overview
Design
The day's trades at a glance: counts, notional and mark-to-market, MTM by desk (click a bar to filter the rest; the drill button goes on to the books), notional by trade date, and the trades behind the numbers.
sample
shipped, read-only
Overview
By product
Time
Now (live)
Data as of 2026-11-06
Bookmarks
Choose one
Click a mark to
filter the page
Export
this page
PDF
PNG
Excel
CSV
{ } JSONL
Subscribe
Trades
15,000
1 row · 3.6 ms · columns
Listening
Notional
3,640,087,648,973
1 row · 2.7 ms · columns
Listening
Mark to market
+3,210,381,877
1 row · 1.8 ms · columns
Listening
Average MTM per trade
+214,025
1 row · 4.2 ms · columns
Listening
MTM by desk (click a bar to filter the page)
DESK-RATES
DESK-MM
DESK-STRUCT
DESK-CREDIT
DESK-SFT
DESK-FI
DESK-EQD
DESK-COMM
DESK-FX
DESK-INFL
4.48bn
401.5m
394.24m
386.06m
26.99m
−30.59m
−85.76m
−395.38m
−681.99m
−1.29bn
10 rows · 2.6 ms · columns
Listening
Notional by trade date
120bn
100bn
80bn
60bn
40bn
20bn
0
129 rows · 3.3 ms · columns · by week
Listening
Trades by desk and currency
DESK
CURRENCY
Trades
Notional
MTM
DESK-COMM CAD 12 99,595,000 +875,931
DESK-COMM EUR 923 1,465,274,740 −2,472,465
DESK-COMM USD 5,055 883,452,132,245 −393,786,717
DESK-CREDIT AUD 81 29,513,556,858 +310,099,789
Currency
Search
Clear
AUD
CAD
CHF



From the command line: `tools/drishti.py pivot run --source var --rows MGMT.DESK --values "var()"` prints the same cube. Details: REPORTS_GUIDE.md, PIVOT_GUIDE.md.

8.9 The Elements gallery

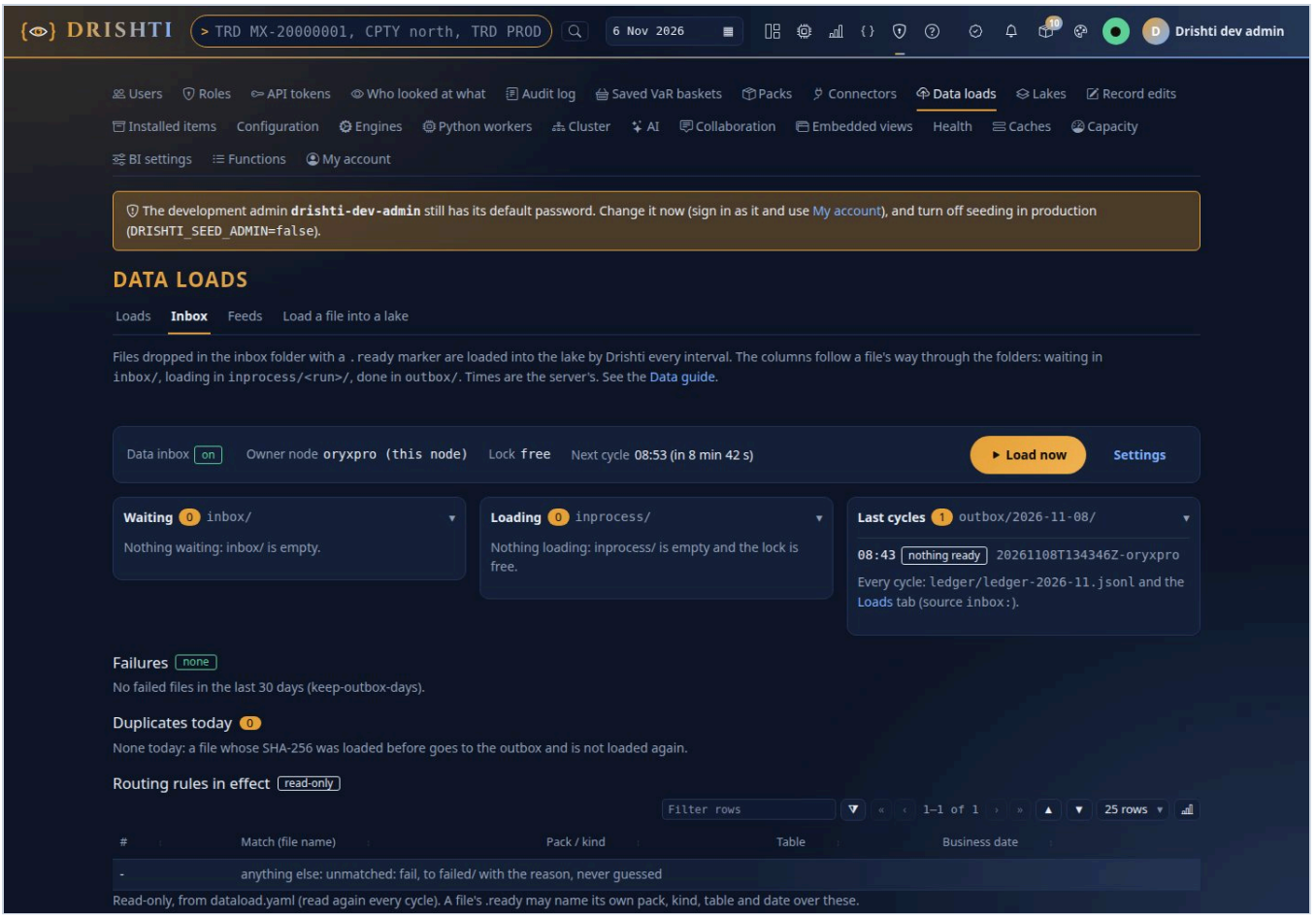
In the UI. Build → Elements gallery (`http://<host>:17480/elements`): every component of the Drishti Elements libraries, live (grid, charts, inputs, maps, the risk widgets); each component's **Playground page** draws it over data you edit in a grid (ELEMENTS.md section 8.4). A component that does not draw here will not draw on a screen either.

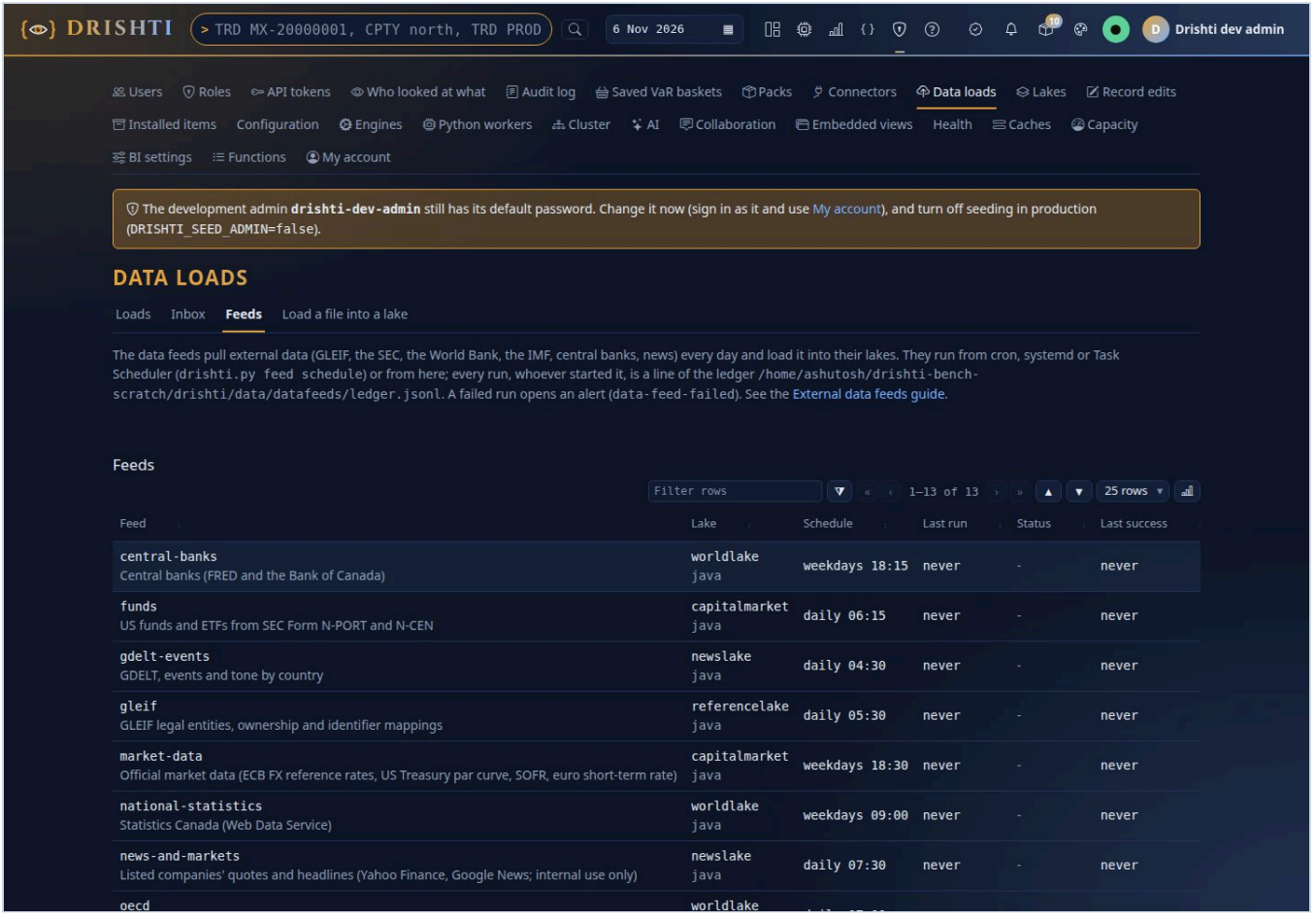


8.10 The data inbox and the feeds

The demo uses only the demo lake: do not drop files into a real lake to test. Check that the inbox runs and what it would take.

In the UI. Admin → Data loads → Inbox: the inbox's folders under `data/dataload/`, this node as its owner, the next cycle, and the files it has seen (none yet). **Admin → Data loads → Feeds**: the public-data feeds with their lake, schedule and licence.


DRISHTI
> TRD MX-20000001, CPTY north, TRD PROD
6 Nov 2026
Drishti dev admin
Users Roles API tokens Who looked at what Audit log Saved VaR baskets Packs Connectors Data loads Lakes Record edits
Installed items Configuration Engines Python workers Cluster AI Collaboration Embedded views Health Caches Capacity
BI settings Functions My account
The development admin drishti-dev-admin still has its default password. Change it now (sign in as it and use My account), and turn off seeding in production (DRISHTI_SEED_ADMIN=false).
DATA LOADS
Loads Inbox Feeds Load a file into a lake
Files dropped in the inbox folder with a .ready marker are loaded into the lake by Drishti every interval. The columns follow a file's way through the folders: waiting in inbox/, loading in inprocess/<run>/, done in outbox/. Times are the server's. See the Data guide.
Data inbox on
Owner node oryxpro (this node)
Lock free
Next cycle 08:53 (in 8 min 42 s)
Load now
Settings
Waiting 0 inbox/
Nothing waiting: inbox/ is empty.
Loading 0 inprocess/
Nothing loading: inprocess/ is empty and the lock is free.
Last cycles 1 outbox/2026-11-08/
08:43 nothing ready 20261108T134346Z-oryxpro
Every cycle: ledger/ledger-2026-11.jsonl and the Loads tab (source inbox:).
Failures none
No failed files in the last 30 days (keep-outbox-days).
Duplicates today 0
None today: a file whose SHA-256 was loaded before goes to the outbox and is not loaded again.
Routing rules in effect read-only
Filter rows
1–1 of 1
25 rows
Match (file name) Pack / kind Table Business date
- anything else: unmatched: fail, to failed/ with the reason, never guessed
Read-only, from dataload.yaml (read again every cycle). A file's .ready may name its own pack, kind, table and date over these.


DRISHTI
> TRD MX-20000001, CPTY north, TRD PROD
6 Nov 2026
Drishti dev admin
Users Roles API tokens Who looked at what Audit log Saved VaR baskets Packs Connectors Data loads Lakes Record edits
Installed items Configuration Engines Python workers Cluster AI Collaboration Embedded views Health Caches Capacity
BI settings Functions My account
The development admin drishti-dev-admin still has its default password. Change it now (sign in as it and use My account), and turn off seeding in production (DRISHTI_SEED_ADMIN=false).
DATA LOADS
Loads Inbox Feeds Load a file into a lake
The data feeds pull external data (GLEIF, the SEC, the World Bank, the IMF, central banks, news) every day and load it into their lakes. They run from cron, systemd or Task Scheduler (drishti.py feed schedule) or from here; every run, whoever started it, is a line of the ledger /home/ashutosh/drishti-bench-scratch/drishti/data/datafeeds/ledger.jsonl. A failed run opens an alert (data-feed-failed). See the External data feeds guide.
Feeds
Filter rows
1–13 of 13
25 rows
Feed Lake Schedule Last run Status Last success
central-banks
Central banks (FRED and the Bank of Canada)
worldlake
java
weekdays 18:15
never
-
never
funds
US funds and ETFs from SEC Form N-PORT and N-CEN
capitalmarket
java
daily 06:15
never
-
never
gdelt-events
GDELT, events and tone by country
newslake
java
daily 04:30
never
-
never
gleif
GLEIF legal entities, ownership and identifier mappings
referencelake
java
daily 05:30
never
-
never
market-data
Official market data (ECB FX reference rates, US Treasury par curve, SOFR, euro short-term rate)
capitalmarket
java
weekdays 18:30
never
-
never
national-statistics
Statistics Canada (Web Data Service)
worldlake
java
weekdays 09:00
never
-
never
news-and-markets
Listed companies' quotes and headlines (Yahoo Finance, Google News; internal use only)
newslake
java
daily 07:30
never
-
never
oecd
worldlake

The `.ready` file releases a dropped file, and may say how to load it. Empty is fine; its keys are `lake`, `pack`, `kind`, `connector`, `table`, `business-date`, `mode` (`replace` or `append`), `key`, `sha256`, `rows`, `new-table`, `hold`, and `folders` for a tar (DATA_LAKE_GUIDE.md › The .ready file, every key):

```
# data/dataload/inbox/trade_endur.2026-11-06.jsonl.gz.ready  (an example: not for the demo lake)
lake: risklake
pack: trading
mode: replace
```

From the command line:

```
$ drishti-console/.venv/bin/python tools/drishti.py feed list
FEED                LAKE           LOAD  SCHEDULE        LICENCE                              USE
DATASETS
central-banks       worldlake      java  weekdays 18:15  FRED API terms (the default series  attribu
tion    central-bank, rate-series
funds               capitalmarket  java  daily 06:15     Public domain (U.S. Government wor  allowed
fund, fund-holder, index-funds
gdelt-events        newslake       java  daily 04:30     GDELT terms of use (free for any u  attribu
tion    event-country
gleif               referencelake  java  daily 05:30     CC0 1.0 (attribution appreciated)   allowed
lei-entity, lei-relationship, lei-mapping
…
world-economy       worldlake      java  daily 06:45     World Bank CC BY 4.0; IMF terms (n  interna
l only  economy, macro-indicator, …
yahoo-prices (off)  capitalmarket  java  weekdays 22:30  Yahoo terms of service (personal u  interna
l only  price-history
$ drishti-console/.venv/bin/python tools/drishti.py feed run gleif --dry-run
dry run: feed gleif (GLEIF legal entities, ownership and identifier mappings), business date 2026-11-
08, window 2026-10-09..2026-11-08, datasets lei-entity, lei-relationship, lei-mapping -> lake referen
celake by java
  dry run: gleif, sources golden, api, mapping
    golden (CC0 1.0 (attribution appreciated); committable): 3 request(s) or file(s), cache <drishti
>/data/datafeeds/sources/cache/gleif
      https://goldencopy.gleif.org/api/v2/golden-copies/publishes/latest
      …
2026-11-08T07:23:11-05:00 feed dry-run exit=0 feed=gleif date=2026-11-08 files=0 rows=0 seconds=0.0
```

A dry run contacts nothing and writes nothing: it says what a run would fetch and load. On an offline machine stop there.

8.11 Modules off and on

Switch the banking module off, restart, check that every risk page is gone, then switch it back on.

In YAML (`config/modules/banking.yaml`): `enabled: false`. **From the command line** instead:

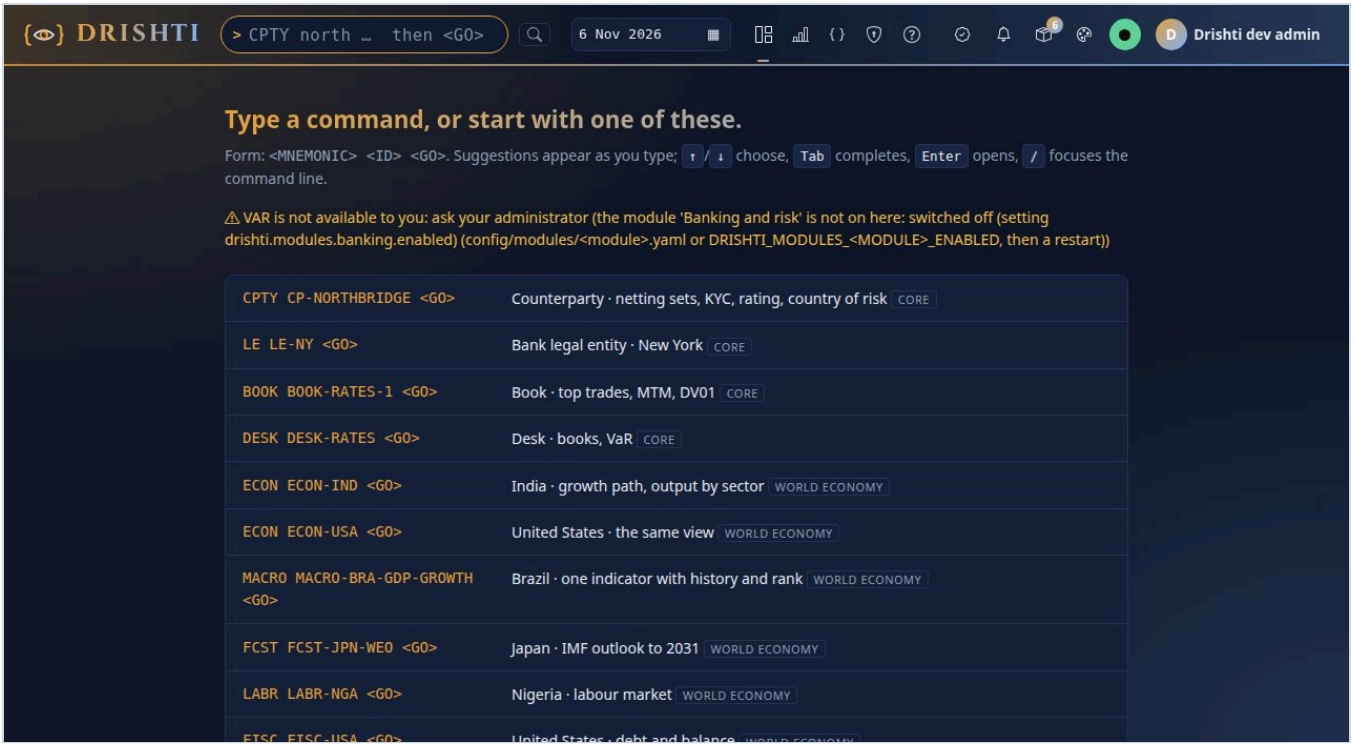
```
curl -s -X POST -H "Authorization: Bearer $ADMIN_TOKEN" -H 'Content-Type: application/json' http://<h
ost>:18480/api/v1/admin/modules \
  -d '{"module": "banking", "enabled": false, "reason": "QA: banking off"}'
```

Or for one run only: `DRISHTI_MODULES_BANKING_ENABLED=false` in the server's environment. Restart the server.

You should see `GET /api/v1/modules` list banking under `off` with its reason; the Risk menu gone from the top bar (its engines are the banking module's); Home offering no Risk or Trading lens; `VAR BANK` answering *VAR is not available to you: ask your administrator (the module 'Banking and risk' is not on here …)*; and the risk API answering `404 DRS-1025`:

```
$ curl -s -H "Authorization: Bearer $ADMIN_TOKEN" "http://<host>:18480/api/v1/risk/var/scope?scope=BANK"
{"type":"about:blank","title":"module off","status":404,"detail":"DRS-1025 the module 'banking' is not on here: switched off (setting drishti.modules.banking.enabled) …","code":"DRS-1025","module":"banking"}
```

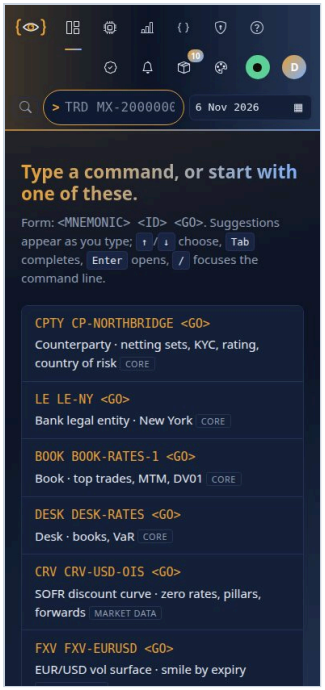
In a cluster switch a module on every node alike (section 10.5).

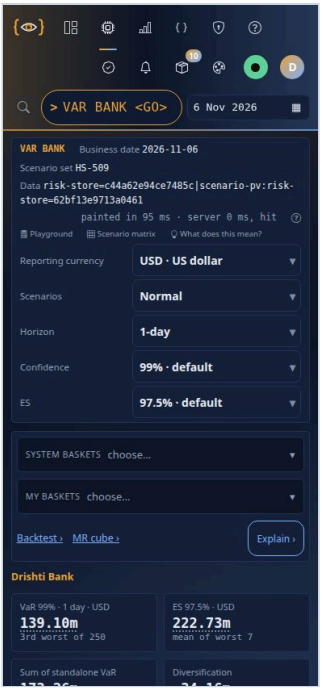


Set `enabled: true` (or remove the variable), restart, and repeat 8.6: VaR answers again.

8.12 The phone layout

Open the console on a phone (or the browser's device toolbar at 390 px wide): Home and every view are one column with no sideways scroll: the top bar wraps, controls and tiles stack, and the function keys become a scrolling row.

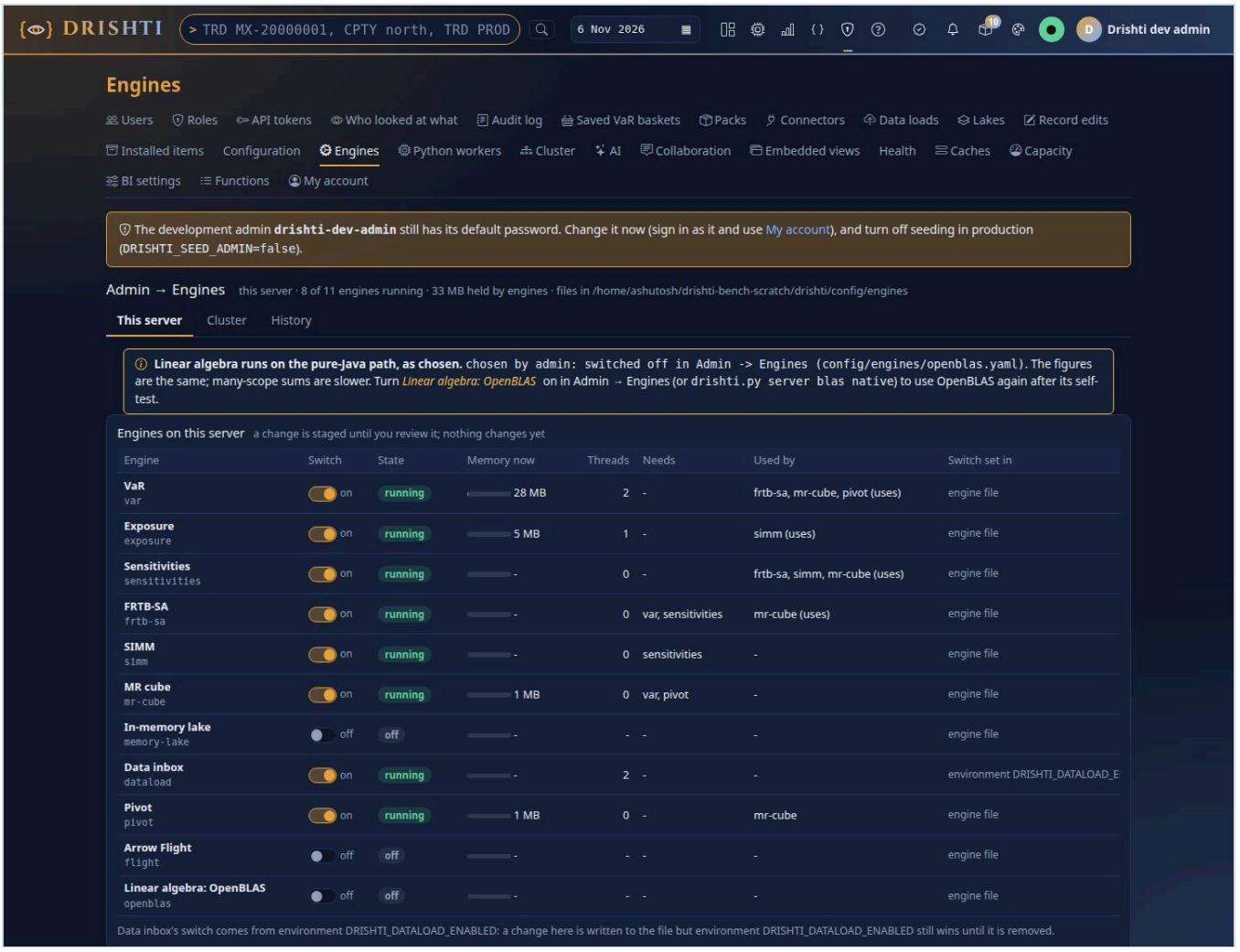




8.13 Logs and metrics

What	Where
The server's log	its standard output (the terminal, `journalctl -u drishti-server`, the scheduled task's `data\logs\server.log`, launchd's `StandardOutPath`); `DRISHTI_LOG_FILE` adds a rolled file
The console's log	its standard output
Metrics	`GET http://<host>:18480/actuator/prometheus` with `Authorization: Bearer $DRISHTI_METRICS_TOKEN` (or an administrator's token); `deploy/prometheus/drishti-alerts.yaml` and `deploy/grafana/drishti-dashboard.json`
Who did what	**Admin → Audit**
The engines on this server	**Admin → Engines**

```
$ curl -s -H "Authorization: Bearer $(cat ~/drishti-token)" http://<host>:18480/actuator/prometheus |
grep -c '^drishti_'
218
```



Every endpoint and alert: OPERATIONS_GUIDE.md › Monitoring and health, the logs: › Logs (docs/guides/OPERATIONS_GUIDE.md#139-logs).

9. The QA checklist

Tick one column per operating system tested. *Expected* is what section 8 shows.

#	Check	Expected	Windows	Linux	macOS
1	`java -version`, Python 3.13, `doctor`	Java 21; `doctor: all clear`	☐	☐	☐
2	Windows only: nothing installed but Java and Python	no `.exe` copied; scripts run with `-ExecutionPolicy Bypass -File`	☐	—	—
3	`lake demo`	`Demo lake ready: … 86 tables, 60 business days …`	☐	☐	☐
4	`lake check`	no error line	☐	☐	☐
5	Server start	`Started DrishtiApplication`, no `ERROR` line	☐	☐	☐
6	`/actuator/health`, console `/readyz`	`"status":"UP"`; `"server":"reachable"`	☐	☐	☐
7	`/api/v1/modules`	banking, capitalmarkets, economy loaded; `off` empty	☐	☐	☐

#	Check	Expected	Windows	Linux	macOS
8	Sign in, change the password, make a token	Home opens; `doctor --token-file` green	☐	☐	☐
9	Home through Risk and through World	each lens's Home with its cards	☐	☐	☐
10	Admin → Lakes	the demo lake's tables with rows and dates	☐	☐	☐
11	`VAR BANK`, `VAR DESK RATES`; Explain ›, Why it moved	the bank's VaR and ES, the scope tree; one desk; the change in steps that add up	☐	☐	☐
12	`EXP BANK`, Pre-deal check	the exposure explorer; the check's form	☐	☐	☐
13	BI Trades overview; `MCUBE`	the report with live slicers; VaR by desk	☐	☐	☐
14	Build → Elements gallery, a Playground page	every component drawn; an edited cell redraws the visual	☐	☐	☐
15	Admin → Data loads → Inbox and Feeds; `feed list`, `feed run gleif --dry-run`	the owner node; the feeds; `exit=0`	☐	☐	☐
16	Banking off, restart	`off` lists banking; no risk pages; `DRS-1025`	☐	☐	☐
17	Banking on, restart	VaR answers again	☐	☐	☐
18	Phone width (390 px)	one column, no sideways scroll	☐	☐	☐
19	Logs and `/actuator/prometheus`	no `ERROR`; `drishti_` metrics	☐	☐	☐
20	Service start at boot, stop with drain	the process comes back after a reboot; a stop drains first	☐	☐	☐
21	Optional: the cluster checklist	section 10.10	☐	☐	☐

10. A cluster of three

Optional. Several servers act as one (**Drishti*net**): each holds its packs and data, they learn of each other by gossip, a request for something one node holds is forwarded to it, and the console moves to another server when its own stops. How it works inside, at depth: CLUSTER_GUIDE.md (the one home for the internals). This section is the complete deployment and QA path.

10.1 When to choose a cluster, and the layout

Choose a cluster when	Stay on one node when
one machine's memory or CPU is not enough for the trades, scenarios and users	one machine holds the day comfortably (the sizing planner says so)
the service must survive the loss of a machine, or be upgraded without downtime	a short stop for an upgrade is acceptable
teams or regions want their own servers with their own packs, seen as one	

The three-node layout used here (and by `deploy/stack/`, the repository's ready-made local cluster):

Node	Role	Packs and engines	Special settings
`drishti-a`	**owner and data-load node**: the seed the others are given, the only node that runs the data inbox	all packs; VaR, exposure, the inbox	`DRISHTI_DATALOAD_ENABLED=true`, `DRISHTI_DATALOAD_OWNER=true`
`drishti-b`	**compute**: VaR shards, exposure, BI	all packs (or a subset)	`DRISHTI_COMPUTE_ENABLED=true`
`drishti-c`	**compute**, the third copy of every shard	the same	the same
console(s)	one per machine, or one behind a load balancer	—	`DRISHTI_BACKEND_URLS=http://<host-a>:18480` (one is enough: it learns the rest)

In production the three are three machines, each with a server and a console, the lake and the identity database shared and highly available outside them (CLUSTER_GUIDE.md › Three machines in production). Larger estates by role: CLUSTER_GUIDE.md › Roles and how many.

10.2 Network and ports

Port	Between	Notes
server port (18480)	console → server, server ↔ server (forwarded requests, digests, seeds)	open between every pair of nodes and from the consoles
gossip port	server ↔ server only	`gossip-port: auto` (default: a free port chosen at start and advertised); **a number** (for example `7946`) when a firewall must name it; `off` shares the server's port. Admin → Health shows `gossipPort`
console port (17480)	browsers → console	from people's networks

Advertise the real address, never `localhost`, `127.0.0.1` or `0.0.0.0`. Each node tells the others where to call it (`advertise-url`); left out, the node uses the default route's interface, which is right on most machines. A loopback or wildcard address is refused by the CLI, Admin → Cluster and the server alike (CLUSTER_GUIDE.md › Real addresses only). Clocks within 60 seconds (NTP): signed calls outside that window are refused.

	Windows	Linux	macOS
the real address	`ipconfig` (the adapter with the default gateway)	`ip route get 1.1.1.1` (`src …`)	`ipconfig getifaddr en0`
open the ports	`New-NetFirewallRule -DisplayName 'Drishti' -Direction Inbound -Protocol TCP -LocalPort 18480,7946 -Action Allow`	`sudo firewall-cmd --add-port={18480,7946}/tcp --permanent && sudo firewall-cmd --reload` (or `ufw allow`)	System Settings → Network → Firewall: allow `java`

10.3 Shared data: the lake and paths

Every node reads the same lake at the same path. A lake's root is in its lake file (`config/lakes/<lake>.yaml`, `root:`), so the same file works on every node only when the path is the same everywhere:

Storage	How	Notes
S3 or an S3-compatible store	the lake file's root `s3://bucket/lakes/demo` and the connector's `s3.*` settings	nothing mounted; the recommended production choice
NFS (Linux, macOS)	mount at the same path on every node, for example `/mnt/drishti/delta`	`DRISHTI_DELTA_ROOT=/mnt/drishti/delta/demo` on each
EFS (AWS)	an access point mounted at the identical path in every task	CLUSTER_GUIDE.md › Drishti on AWS
SMB (Windows)	a UNC path, the same on every node: `\\fileserver\drishti\delta\demo`	the service account needs read access; a mapped drive letter is per user, so use the UNC path

What is not shared: each node's own data folder (`DRISHTI_DATA_DIR`: caches, hot files, its net state), which it rebuilds from the lake. **Shared besides the lake:** the identity database (PostgreSQL in a production cluster, never the SQLite default) and the content store (`config/store/store.yaml`, the same on every node). Loads are announced to every node, so all read the same lake version.

10.4 Configuration per node

Each node has its **own** `config/net/` folder (`DRISHTI_NET_DIR`) and data folder; everything else (packs, lakes, modules, lenses, engines) is the **same** on every node. On each node:

```
# config/net/node.yaml on drishti-b
enabled: true
node-id: drishti-b
advertise-url: http://<host-b>:18480        # left out: this machine's real address
gossip-port: auto                           # or 7946 behind a firewall
secret: ${DRISHTI_NET_SECRET}               # a reference: the key itself never goes in a file
```

```
# config/net/peers/drishti-a.yaml on drishti-b and drishti-c: the seed
url: http://<host-a>:18480
```

From the command line the same, checked as the server would:

```
export DRISHTI_NET_SECRET=<the cluster key>
drishti-console/.venv/bin/python tools/drishti.py net set node.node-id drishti-b
drishti-console/.venv/bin/python tools/drishti.py net set node.secret '${DRISHTI_NET_SECRET}'
drishti-console/.venv/bin/python tools/drishti.py net set node.advertise-url http://<host-b>:18480
drishti-console/.venv/bin/python tools/drishti.py net set node.enabled true
drishti-console/.venv/bin/python tools/drishti.py net add-peer drishti-a <host-a>:18480
drishti-console/.venv/bin/python tools/drishti.py net check                # exit 0: every file woul
d be taken
```

In the UI (once a server runs): **Admin → Cluster → Files** (open `node.yaml`, edit, **Check**, **Save**), **Admin → Cluster → Members → Add a seed**.

The environment per node (the example layout of `deploy/stack/stack.yaml`, the repository's three-node cluster on one machine):

Variable	drishti-a	drishti-b	drishti-c
`DRISHTI_NET_SECRET`, `DRISHTI_TOKEN_SECRET`, `DRISHTI_SESSION_SECRET`	the same three values on every node	same	same
`DRISHTI_NET_DIR`, `DRISHTI_DATA_DIR`	its own	its own	its own
`DRISHTI_DELTA_ROOT` (or the lake files' roots)	the shared lake path	the same	the same
`DRISHTI_DATALOAD_ENABLED`, `DRISHTI_DATALOAD_OWNER`	`true`, `true`	unset	unset
`DRISHTI_COMPUTE_ENABLED`	`true`	`true`	`true`
`DRISHTI_PACKS`, `DRISHTI_LENSES`, `config/modules/*`	the same everywhere	same	same

10.5 Secrets

The three secrets (the cluster key, the token secret, the session secret) are the same on every node and the consoles, made once with `openssl rand -hex 32`, kept where only the service account reads them (the environment file, the vault, Secrets Manager), never in a file under `config/`. A file with a key in clear is refused, and Admin → Cluster shows only whether the key is set. Rolling the key without a stop: CLUSTER_GUIDE.md › Security. **Modules and licence:** every node reads its own `config/modules/` and `config/access/licence.yaml`: give each node the same files (from the same release folder), so the same modules are on everywhere; a module on one node and off on another leaves its pages working only through that node.

10.6 Start order and joining

1. The shared pieces first: the lake reachable, the identity database (and Kafka, if used).
2. The seed, `drishti-a`; then `drishti-b` and `drishti-c` (any order, at once is fine).
3. The consoles last, each given the entry node (or several); each learns the other members from the cluster.

Each node starts with `bin/start-server --node` (section 6.1 has every option). All nodes share one env-file with the three secrets (6.4), here `cluster.env`. Each node advertises its machine's real address (found by the script; `--advertise` names another), never `127.0.0.1`.

Three machines (the production layout): `drishti-a` on `<host-a>`, `drishti-b` on `<host-b>`, `drishti-c` on `<host-c>`, each on port 18480 with its own data folder. Linux and macOS:

```
# on <host-a>: the seed
bin/start-server.sh --node drishti-a --env-file /etc/drishti/cluster.env --background
# on <host-b> and <host-c>: join through the seed
bin/start-server.sh --node drishti-b --seed <host-a>:18480 --env-file /etc/drishti/cluster.env --back
ground
bin/start-server.sh --node drishti-c --seed <host-a>:18480 --env-file /etc/drishti/cluster.env --back
ground
# the console (on any machine): given the entry node, it learns drishti-b and drishti-c
bin/start-console.sh --server http://<host-a>:18480 --env-file /etc/drishti/cluster.env --background
```

Windows (each line on its machine):

```
bin\start-server.cmd --node drishti-a --env-file C:\drishti-secrets\cluster.env --background
bin\start-server.cmd --node drishti-b --seed <host-a>:18480 --env-file C:\drishti-secrets\cluster.env
--background
bin\start-server.cmd --node drishti-c --seed <host-a>:18480 --env-file C:\drishti-secrets\cluster.env
--background
bin\start-console.cmd --server http://<host-a>:18480 --env-file C:\drishti-secrets\cluster.env --back
ground
```

The first start of a node makes its Drishti*net folder, `<data>/nodes/<node>/net` (a `node.yaml` naming `${DRISHTI_ADVERTISE_HOST}`, `${DRISHTI_PORT}` and `${DRISHTI_NET_SECRET}`, never the values), and `--seed` writes `peers/seed-<host>-18480.yaml`. A folder made and checked by hand (10.4) is given with `--node-dir`: `bin/start-server.sh --node drishti-b --node-dir /etc/drishti/net …`. A console may be given several entry nodes, `--server http://<host-a>:18480,http://<host-b>:18480`: it uses the first that answers.

Three nodes on one machine (a laptop, a QA box): different ports, and each node its own data folder (`--node-data`; the script refuses two nodes on one), while the pid files, logs and temporary folders stay in the one data folder, named after the node:

```
bin/start-server.sh --node drishti-a --port 18480 --node-data data/nodes/drishti-a/data --env-file cl
uster.env --background
bin/start-server.sh --node drishti-b --port 18481 --node-data data/nodes/drishti-b/data --seed <host
>:18480 --env-file cluster.env --background
bin/start-server.sh --node drishti-c --port 18482 --node-data data/nodes/drishti-c/data --seed <host
>:18480 --env-file cluster.env --background
bin/start-console.sh --server http://<host>:18480 --env-file cluster.env --background
bin/status.sh
```

On Windows the same lines with `bin\start-server.cmd` and `bin\start-console.cmd`. **You should see** (a run on test ports):

```
t-b: up on http://192.168.2.77:28791
t-c: up on http://192.168.2.77:28792
console: up on http://192.168.2.77:27480
process                    pid  state      port   health     address
server-t-a              1977639  running    28480  UP         http://192.168.2.77:28480
server-t-b              1978776  running    28791  UP         http://192.168.2.77:28791
server-t-c              1979410  running    28792  UP         http://192.168.2.77:28792
console                 1980320  running    27480  UP         http://192.168.2.77:27480
```

and in each server's log:

```
INFO  … GossipConnector : net: gossip port 36009 open (8 platform threads of its own; gossip, probes
and digests only)
INFO  … NetService      : net: node drishti-b joined Drishti*net at http://<host-b>:18480 (incarnatio
n 1, seeds [http://<host-a>:18480], 0 saved members, gossip on port 36009)
INFO  … NetService      : net: member drishti-a joined
INFO  … NetService      : net: member drishti-c joined
```

The console's saved list (`data/console/net/servers.json`) then holds the entry node as `configured` and the other two as `discovered`.

Stopping one node, then every node of the machine (each by its pid file; drain first in production, 10.8):

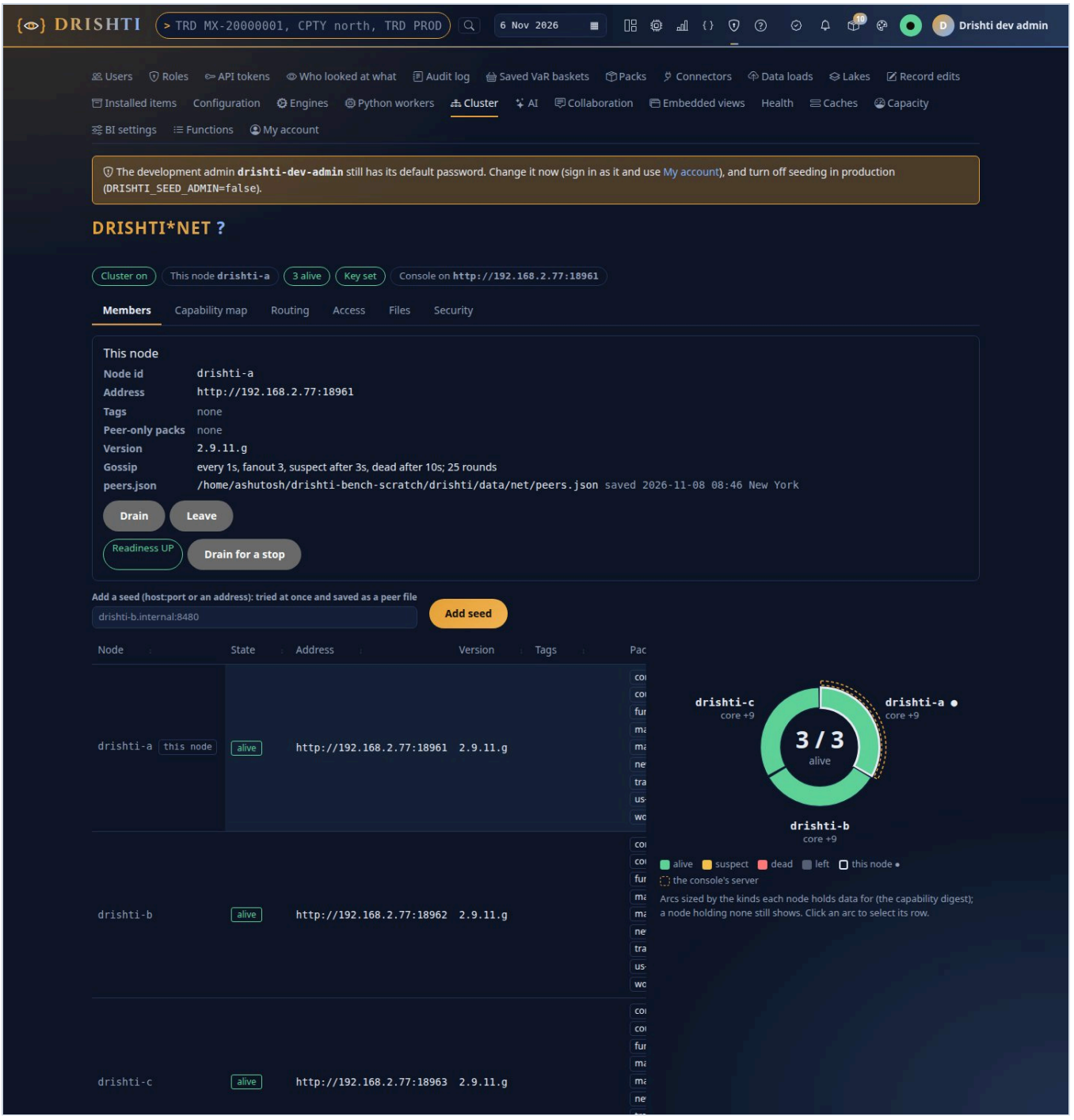
```
bin/stop-server.sh --node drishti-c          # Windows: bin\stop-server.cmd --node drishti-c
bin/stop-console.sh
bin/stop-server.sh --all-nodes               # or bin/stop-all.sh --all-nodes: the console, then ever
y node
```

```
t-c: stopped (pid 1979410)
console: stopped (pid 1980320)
t-a: stopped (pid 1977639)
t-b: stopped (pid 1978776)
```

The repository's own three on one machine. `bin/start-cluster.sh` starts three servers and a console from `deploy/stack/stack.yaml` through these same scripts, with the real address and the secrets made for you; `status.sh` shows them, `stop.sh` stops them (deploy/stack/README.md). The same scripts run on Windows (`bin\start-cluster.cmd`).

10.7 Verify the cluster, step by step

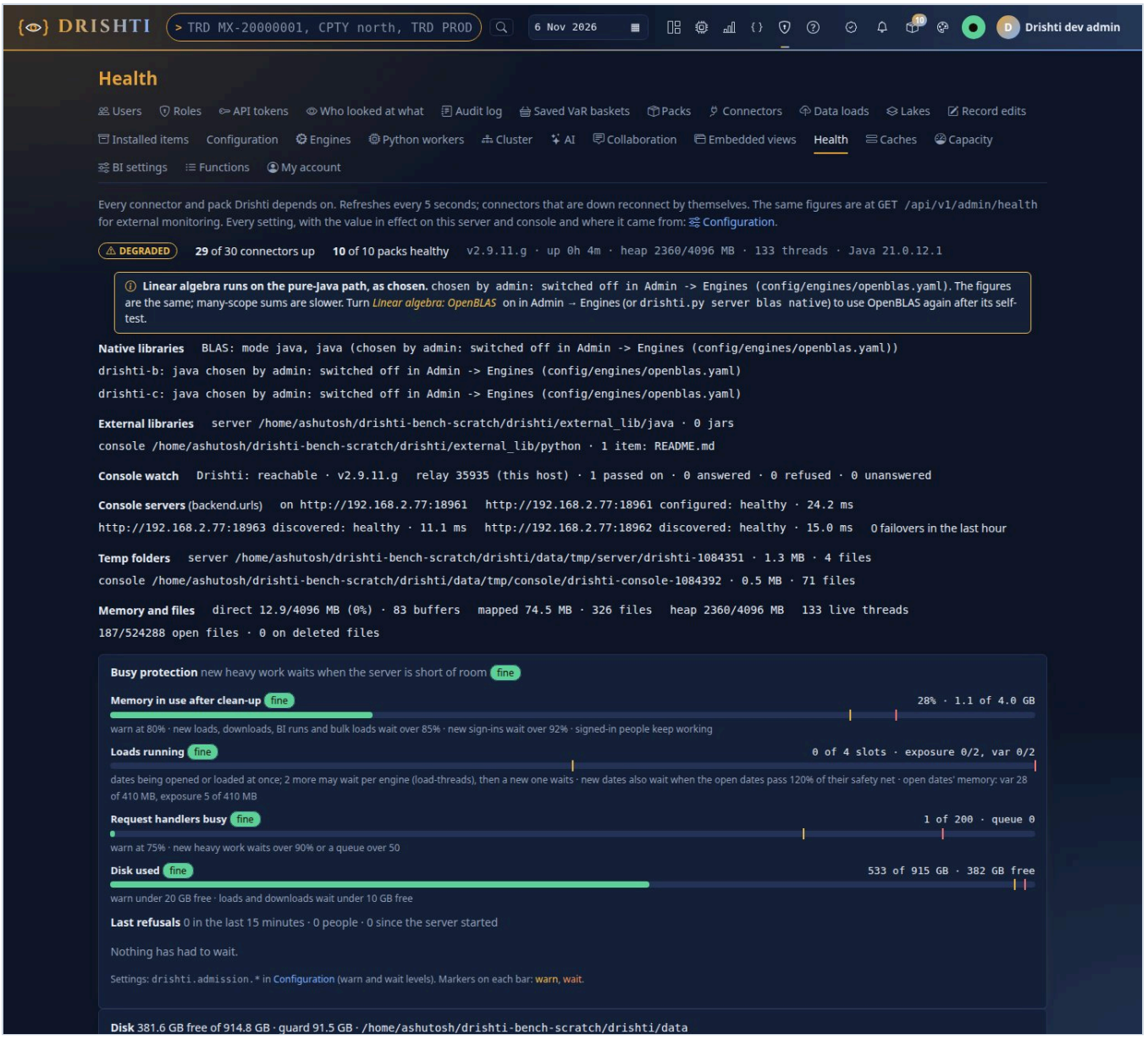
1 · The members. Admin → Cluster (`/admin/net`, the Drishti*net page) → **Members** on any node lists the three `alive`, each with its version, tags, packs and latency; the hash ring is green.



From the command line:

```
$ drishti-console/.venv/bin/python tools/drishti.py net members --server http://<host-a>:18480 --toke
n-file ~/drishti-token
3 member(s); this node drishti-a
  node                  state  address                 version   tags  latency  incarnation
  drishti-a (this node) alive  http://<host-a>:18480   2.9.11.g        -        1
  drishti-b             alive  http://<host-b>:18480   2.9.11.g        4.0 ms   1
  drishti-c             alive  http://<host-c>:18480   2.9.11.g        4.1 ms   1
```

2 · Gossip health. Admin → Health on each node: the Drishti*net line (members alive and suspect, requests forwarded and failed, `gossipPort`).



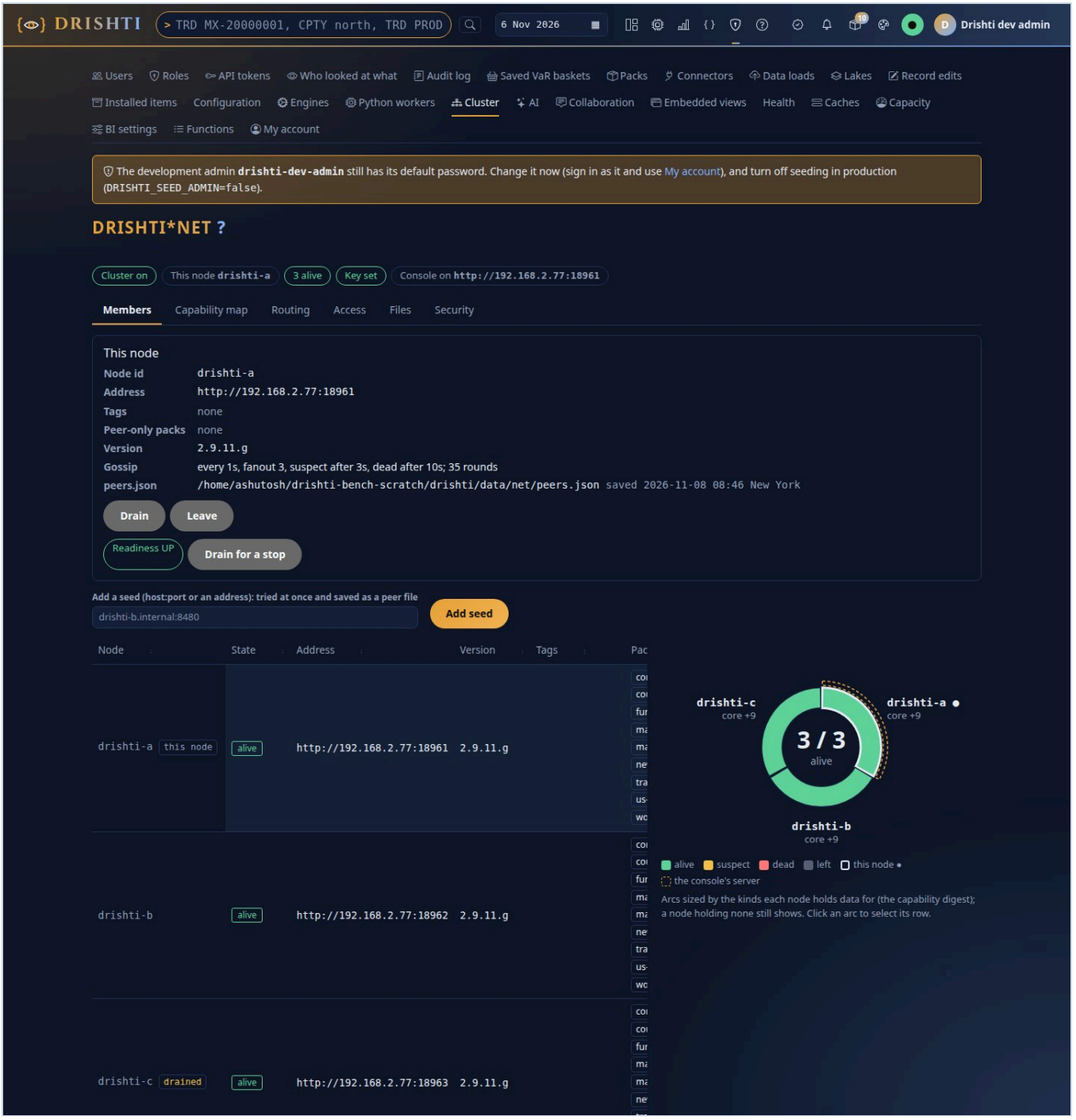
3 · Drain a node, and an answer still comes. Drain one node (as for an upgrade): **Admin → Cluster → Drain for a stop** on that node, or

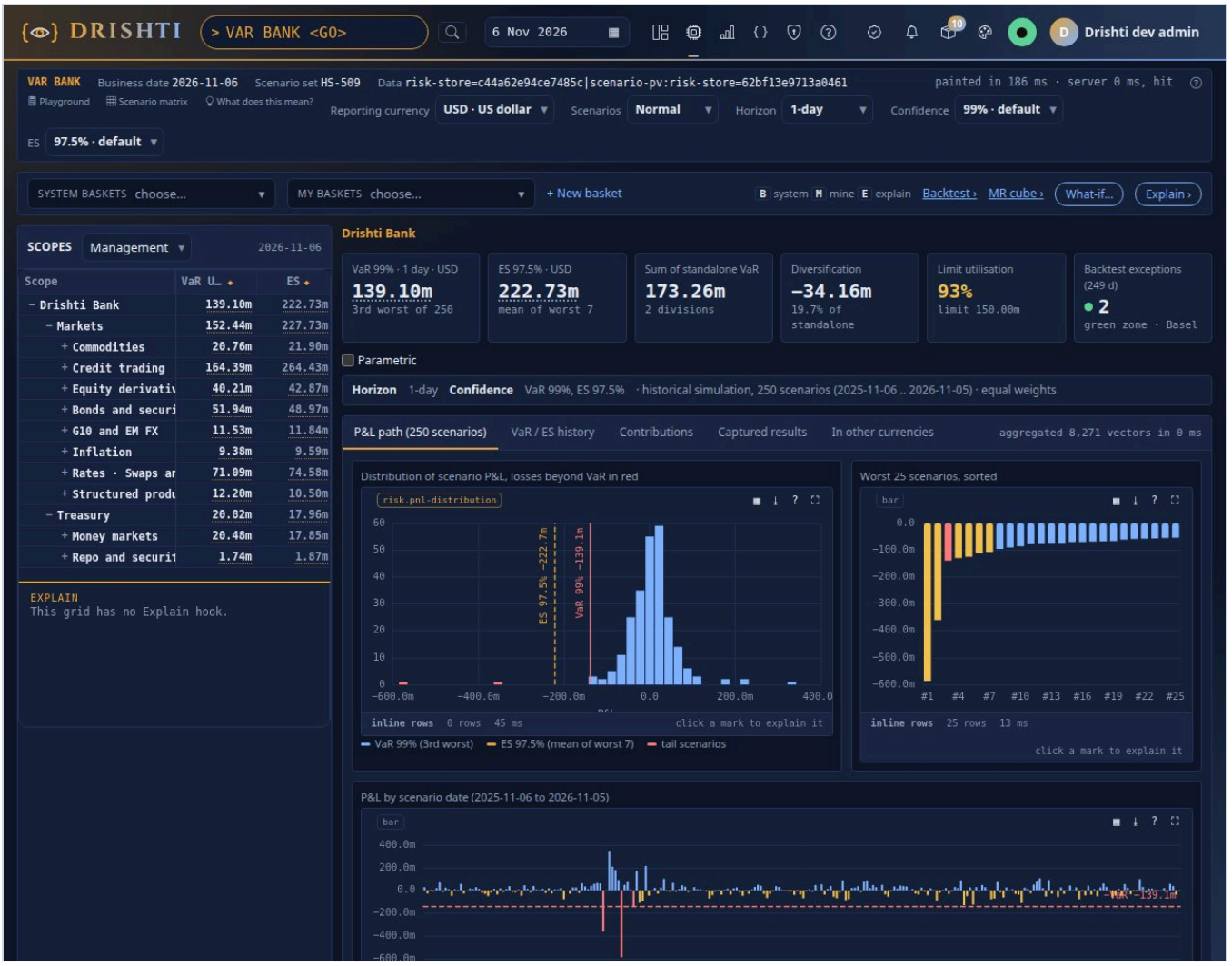
```
curl -s -X POST -H "Authorization: Bearer $ADMIN_TOKEN" -H 'Content-Type: application/json' \
  http://<host-c>:18480/api/v1/admin/node/drain -d '{"on": true}'
curl -s -H "Authorization: Bearer $ADMIN_TOKEN" http://<host-c>:18480/api/v1/admin/node       # "saf
eToStop": true once handed over
```

(Windows: `bin\drishti-node.ps1 drain`; Linux: `bin/drishti-node.sh drain`; a personal token needs the `node:drain` scope.) The drain answers at once with what it is handing over; `safeToStop` turns true within a few seconds:

```
{"node":"drishti-c","drained":true,"since":"2026-11-08T12:59:44Z","safeToStop":false,
 "parts":[{"name":"net","draining":true,"detail":"drained: waiting 5 s for every peer to hear it","sa
feToStop":false},
         {"name":"var-shards","draining":false,"detail":"VaR sharding is off on this server: no shar
ds to hand over","safeToStop":true}],
 "readiness":{"status":"OUT_OF_SERVICE","enabled":true,"waitingFor":{}}}
```

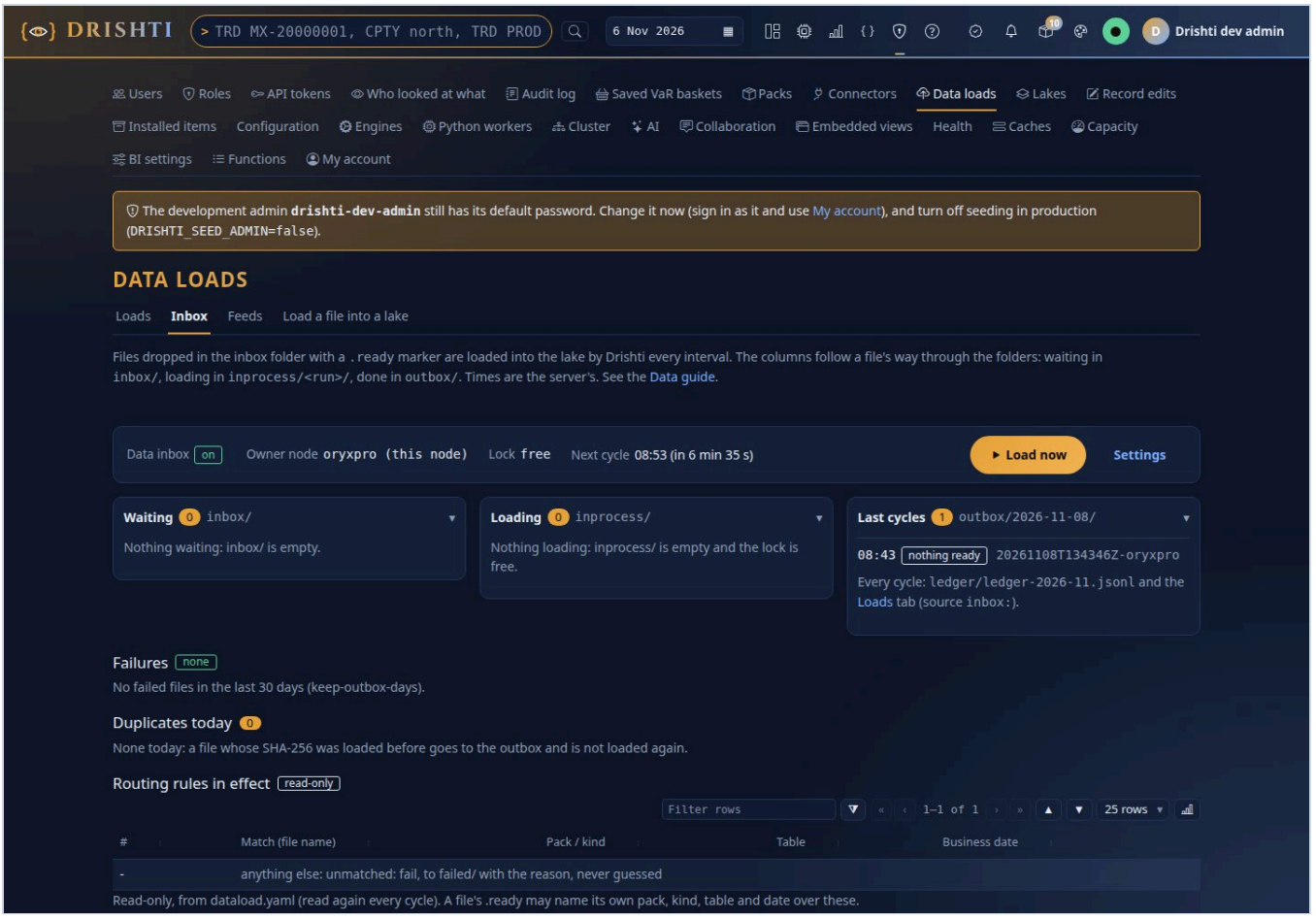
You should see the node marked drained on Admin → Cluster, and `VAR BANK` still answer, from the other nodes:





Undrain with `{"on": false}` (or restart the node).

4 · The inbox owner. Admin → Data loads → Inbox names the owner node, `drishti-a`; on the other nodes the inbox does not run (a folder lock keeps a second runner out anyway).



5 · The same modules everywhere. Ask each node; the answers must be the same:

```
for h in <host-a> <host-b> <host-c>; do
  curl -s -H "Authorization: Bearer $(cat ~/drishti-token)" http://$h:18480/api/v1/modules | python3
-c 'import json,sys; d=json.load(sys.stdin); print([m["id"] for m in d["modules"]], d["off"])'
done
```

6 · The console moves. With the console given `drishti-a` alone, stop `drishti-a`: the open page carries on from another node within a second, still signed in (Admin → Health, *Console servers*, shows the move).

10.8 Rolling upgrade and draining

One node at a time: drain → `safeToStop` → stop → replace the jar → start → readiness → the next node. Consoles first, then the compute nodes, then the owner. `tools/rolling-upgrade.sh` (Windows: `tools\windows\rolling-upgrade.ps1`) does exactly this from an inventory (`deploy/rolling.example.yaml`); `--dry-run` prints every step. The stop hooks of systemd and Docker drain by themselves; on Windows run `drishti-node.ps1 drain -Wait 240` before stopping the task. Details: CLUSTER_GUIDE.md › Zero downtime.

10.9 Per operating system

	Windows	Linux	macOS
start a node	`bin\start-server.cmd --node …` (a scheduled task per node, section 7)	`bin/start-server.sh --node …`; systemd, one unit per node	`bin/start-server.sh --node …`; launchd, one plist per node

	Windows	Linux	macOS
drain before a stop	`bin\drishti-node.ps1 drain -Wait 240`	`ExecStop=` in the unit does it	`bin/drishti-node.sh drain` before `launchctl unload`
the shared lake	SMB, a UNC path	NFS or S3	NFS or S3
three on one machine	three `bin\start-server.cmd --node … --port … --node-data …`	the same with `bin/start-server.sh`, or `bin/start-cluster.sh`	the same as Linux

10.10 The cluster QA checklist

#	Check	Expected	☐
C1	`net check` on each node	exit 0	☐
C2	Each node's log	`joined Drishti*net at http://<real address>` and `member … joined` for the others	☐
C3	Admin → Cluster → Members	three `alive`; addresses are real, none `localhost`	☐
C4	`net members`	three lines, one marked this node	☐
C5	Admin → Health	the Drishti*net line green; `gossipPort` as configured	☐
C6	Drain a node	drained on Admin → Cluster; `safeToStop` true; `VAR BANK` answers	☐
C7	Undrain	the node serves again	☐
C8	Inbox	owner `drishti-a` only	☐
C9	`/api/v1/modules` on each node	identical	☐
C10	Stop the console's server	the page carries on, still signed in	☐
C11	Rolling upgrade `--dry-run`	every step listed, in order	☐

10.11 Cluster troubleshooting

What you see	Do
A node lists no members	its log's `joined … seeds [...]` line; the seed's address and port, the firewall, `enabled: true` with the key set
`node.yaml was not written: '… 127.0.0.1 …' is a loopback or wildcard address`	the machine's real address (the message names it), or leave `advertise-url` out
`401 DRS-5051 cluster credential refused: bad signature`	the same `DRISHTI_NET_SECRET` on every node
`… signed 75 s away from this node's clock`	NTP on every node
A member flaps between `alive` and `suspect`	`gossip-port` on `auto` or a number (not `off`); the firewall must allow it between the nodes
`net: the gossip port … cannot be reached … but … answers`	open the numbered gossip port between the nodes
`503 DRS-5215` naming a shard	an owner is missing: start it; the shard is rebuilt after `stable-seconds`
`409 DRS-5216`	members read different lake versions: the same lake path everywhere; announce the load

What you see	Do
Admin → Cluster warns that the nodes run different editions	the same `DRISHTI_EDITION` on every node, then restart

The rest: CLUSTER_GUIDE.md › When something is wrong.

---

11. Offline QA with your own trade files

A demo or QA rehearsal on a machine **without internet access** (pip and Maven through the company's mirror; on Windows no `.exe` placed on the machine), showing ENDUR trades from the bank's own JSON Lines files beside the synthetic CALYPSO and MUREX trades and all the supporting data, the committed quotes-and-news snapshot (no website is contacted) and the world economy data. Do it the day before the demo. The commands are PowerShell lines; on Linux and macOS the same commands run with `drishti-console/.venv/bin/python` and `/` paths.

Your real trade files are confidential. Keep them **outside** the release folder (below `<private>`, for example `C:\Users\<you>\drishti-private\endur\`); never copy them into the repository. Drishti reads them where they are. Every command is one PowerShell line, after once:

```
$py = "<drishti>\drishti-console\.venv\Scripts\python.exe"; $tools = "<drishti>\tools"
```

11.1 Set up once

1. **The machine:** sections 2 and 3 (Java 21, Python 3.13 and the console's environment). pip reads `%APPDATA%\pip\pip.ini`, Maven `%USERPROFILE%\.m2\settings.xml` for the mirror.
2. **Your files** go in `<private>`, **one per business date**, named by the load rules (DATA_LAKE_GUIDE.md › Routing, business dates, modes): `trade_endur.2026-10-16.jsonl` (or `.jsonl.gz`, `.jsonl.zip`, or a tar of `trade_endur\2026-10-16\` folders). A file named only by its date is refused, with the name to give it.
3. **Two IDE projects, never one folder** (when you use the IDEs): IntelliJ opens `<drishti>` (its run configurations in `run-configs\`: **Drishti build server jar**, **Drishti server (demo lakes)**, which run module `drishti-app` with the class path in an argument file, so Windows' command-line limit raises no warning); PyCharm opens `<drishti>\drishti-console` (`drishti-console\run-configs\`: **Drishti console**, **Build demo lakes**, **Load ENDUR file into the lake**), with the `.venv` interpreter (OPERATIONS_GUIDE.md › PyCharm). Opened on the same folder they overwrite each other's `.idea` (IntelliJ then says **module not specified**: close both, delete `<drishti>\.idea`, reopen).

11.2 Put the data in place

One command for everything (stop the server first). `build_demo_lakes.py` builds every demo lake over the sixty business days of the committed test data, moved to end on the date of your newest file, then replaces the synthetic ENDUR trades with your files:

```
& $py "$tools\build_demo_lakes.py" --endur-dir <private> --dry-run     # prints every step, changes n
othing
& $py "$tools\build_demo_lakes.py" --endur-dir <private>
```

You should see `1/3 the demo lakes …` (15 to 45 minutes; `data\delta\demo` ends at about 3.4 GB), then `2/3 your ENDUR files (N: <first date> … <last date>)` with one `lake-build ok … rejected=0` line per file, then `3/3 done in … s`. A line every 30 seconds says which step runs and which table is being written; *nothing written for N s* means it is waiting (a slow disk, or antivirus scanning `data`).

You want	Add
synthetic data only, ending today	leave out `--endur-dir`
the synthetic ENDUR trades kept beside yours	`--keep-synthetic-endur`
your last date	`--as-of 2026-10-16`
only your files loaded again	`--endur-dir <private> --files-only` (no server restart needed)
another system's files	`--source-dir <folder> --system calypso --files-only`
some datasets only	`--only trades/calypso,trades/murex --as-of <the lakes' last date>`
one system in Iceberg, the others in Delta	`--store endur=delta --store calypso=iceberg` (`--store all=iceberg` for every system)
your bank's names instead of the demo's	edit `config\demo\names.yaml`, then `--only core/legal-entity,market-risk/pnl-history --as-of <date>` (DATA_GUIDE.md › Your bank's names)
a running server told at once	`--server http://<host>:18480` with a `loads:write` token in `DRISHTI_TOKEN` (or `--token-file`)

By hand, one file:

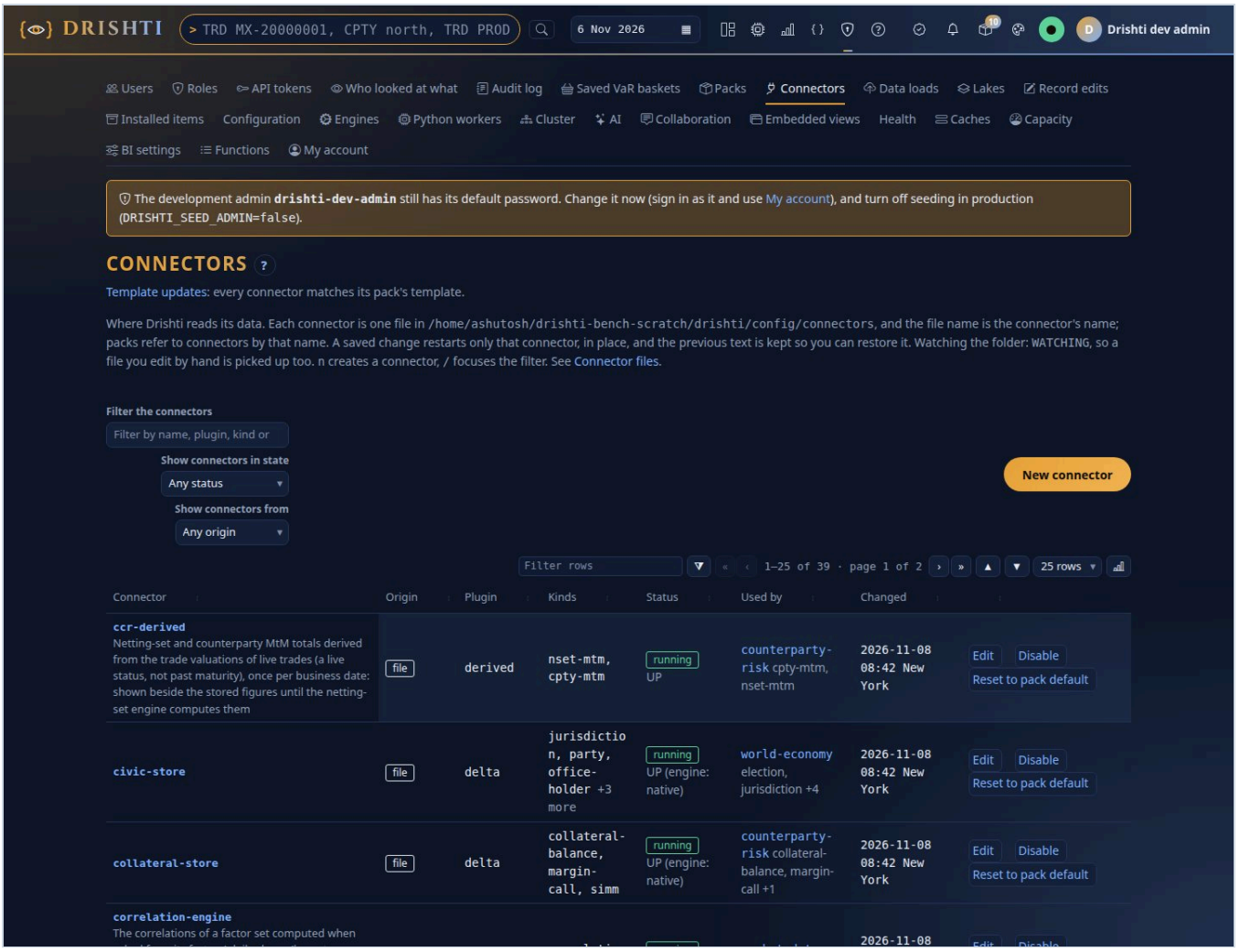
```
& $py "$tools\drishti.py" lake build --from <private>\trade_endur.2026-10-15.jsonl --pack <drishti>\config\packs\banking\trading --connector trade-endur --key TRADE_ID --mode overwrite-dates --lake demo
```

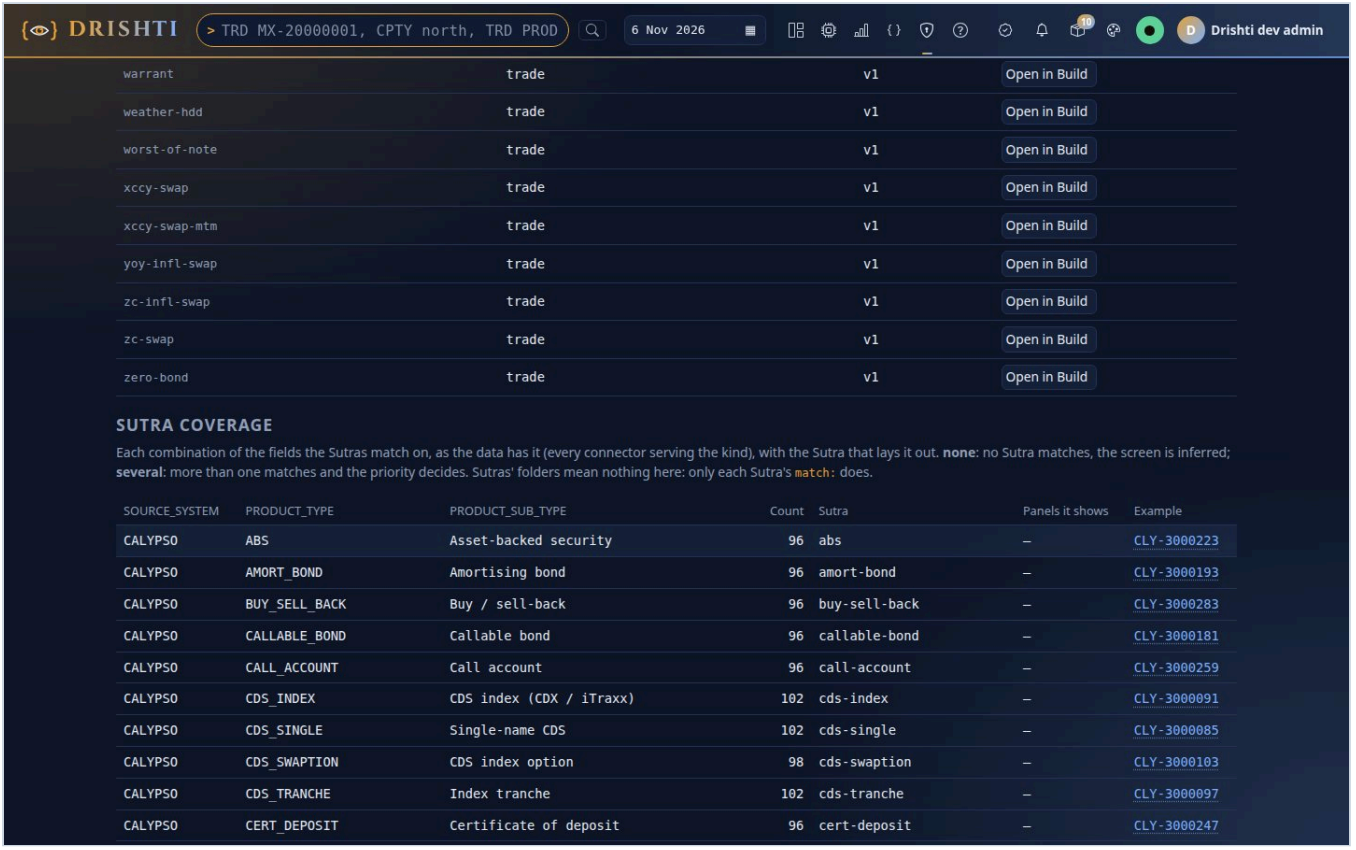
The business date comes from the file name; the pack's schema checks each trade; `--mode overwrite-dates` makes a repeat safe. The last line reads `lake-build ok exit=0 … kind=trade … rows=500 rejected=0 dates=1 …`; a non-zero `rejected` is explained, with the line, above it. Use full paths: PyCharm's terminal opens in `drishti-console`, where a relative path makes a stray `drishti-console\data`. With the Java loader (every core, bounded memory) add `--engine java` (CLI_REFERENCE.md › The parallel lake loader).

11.3 Start and check

Start the server with `bin/start-server.sh --demo` (Windows `bin\start-server.cmd --demo`) and the console as in section 6 (or the IntelliJ configuration **Drishti server (demo lakes)** and the PyCharm one **Drishti console**; never the ▶ in the margin of `DrishtiApplication.java`, which sets none of the demo settings). Then:

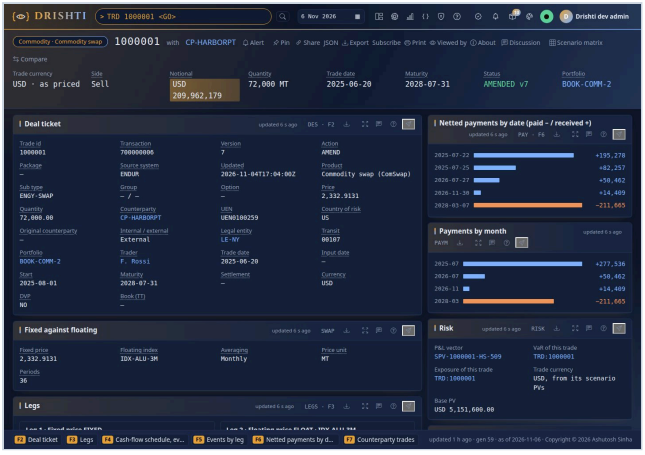
- **Admin → Lakes**: `trading/trade_endur` has **your** rows and dates.
- **Admin → Connectors**: `trade-endur`, `trade-calypso`, `trade-murex` and the stores RUNNING; the live quote and news connectors disabled and the news store serving the snapshot.
- **Admin → Health** green.
- **Admin → Packs → trading → Sutra coverage**: every SOURCE_SYSTEM × PRODUCT_TYPE of your data has a Sutra (a row without one is a product type the pack does not know yet, laid out by inference).


DRISHTI
> TRD MX-20000001, CPTY north, TRD PROD
6 Nov 2026
Drishti dev admin
Users Roles API tokens Who looked at what Audit log Saved VaR baskets Packs Connectors Data loads Lakes Record edits
Installed items Configuration Engines Python workers Cluster AI Collaboration Embedded views Health Caches Capacity
BI settings Functions My account
The development admin drishti-dev-admin still has its default password. Change it now (sign in as it and use My account), and turn off seeding in production (DRISHTI_SEED_ADMIN=false).
CONNECTORS
Template updates: every connector matches its pack's template.
Where Drishti reads its data. Each connector is one file in /home/ashutosh/drishti-bench-scratch/drishti/config/connectors, and the file name is the connector's name; packs refer to connectors by that name. A saved change restarts only that connector, in place, and the previous text is kept so you can restore it. Watching the folder: WATCHING, so a file you edit by hand is picked up too. n creates a connector, / focuses the filter. See Connector files.
Filter the connectors
Filter by name, plugin, kind or
Show connectors in state
Any status
Show connectors from
Any origin
New connector
Filter rows
1–25 of 39 · page 1 of 2
25 rows
Connector Origin Plugin Kinds Status Used by Changed
ccr-derived
Netting-set and counterparty MtM totals derived from the trade valuations of live trades (a live status, not past maturity), once per business date: shown beside the stored figures until the netting-set engine computes them
file derived nset-mtm, cpty-mtm running UP counterparty-risk cpty-mtm, nset-mtm 2026-11-08 08:42 New York Edit Disable Reset to pack default
civic-store
file delta jurisdiction, party, office-holder +3 more running UP (engine: native) world-economy election, jurisdiction +4 2026-11-08 08:42 New York Edit Disable Reset to pack default
collateral-store
file delta collateral-balance, margin-call, simm running UP (engine: native) counterparty-risk collateral-balance, margin-call +1 2026-11-08 08:42 New York Edit Disable Reset to pack default
correlation-engine
The correlations of a factor set computed when

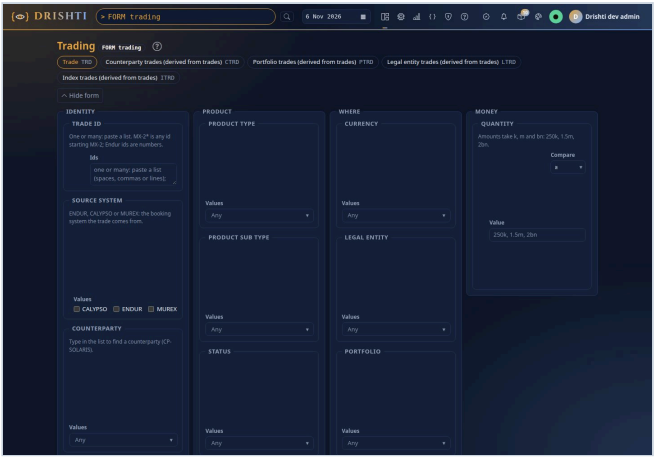

DRISHTI
> TRD MX-20000001, CPTY north, TRD PROD
6 Nov 2026
Drishti dev admin
warrant trade v1 Open in Build
weather-hdd trade v1 Open in Build
worst-of-note trade v1 Open in Build
xccy-swap trade v1 Open in Build
xccy-swap-mtm trade v1 Open in Build
yoy-infl-swap trade v1 Open in Build
zc-infl-swap trade v1 Open in Build
zc-swap trade v1 Open in Build
zero-bond trade v1 Open in Build
SUTRA COVERAGE
Each combination of the fields the Sutras match on, as the data has it (every connector serving the kind), with the Sutra that lays it out. none: no Sutra matches, the screen is inferred; several: more than one matches and the priority decides. Sutras' folders mean nothing here: only each Sutra's match: does.
SOURCE_SYSTEM PRODUCT_TYPE PRODUCT_SUB_TYPE Count Sutra Panels it shows Example
CALYPSO ABS Asset-backed security 96 abs – CLY-3000223
CALYPSO AMORT_BOND Amortising bond 96 amort-bond – CLY-3000193
CALYPSO BUY_SELL_BACK Buy / sell-back 96 buy-sell-back – CLY-3000283
CALYPSO CALLABLE_BOND Callable bond 96 callable-bond – CLY-3000181
CALYPSO CALL_ACCOUNT Call account 96 call-account – CLY-3000259
CALYPSO CDS_INDEX CDS index (CDX / iTraxx) 102 cds-index – CLY-3000091
CALYPSO CDS_SINGLE Single-name CDS 102 cds-single – CLY-3000085
CALYPSO CDS_SWAPTION CDS index option 98 cds-swaption – CLY-3000103
CALYPSO CDS_TRANCHE Index tranche 102 cds-tranche – CLY-3000097
CALYPSO CERT_DEPOSIT Certificate of deposit 96 cert-deposit – CLY-3000247

11.4 The demo, page by page

#	Show	How
1	An ENDUR deal from your file	type one of your trade ids → Enter: ticket, legs, cash flows, resets, events
2	Trades by system and product	**Views → Forms → Trading**: Source system `ENDUR`, a product type → Search; then `CALYPSO`, `MUREX`
3	A CALYPSO and a MUREX deal	`/v/trade/CLY-3000157`, `/v/trade/MX-20000001`
4	Counterparty → netting set → CSA → collateral	`CTRD CP-NORTHPLAINS`, then click through
5	A curve	`/v/commodity-curve/CMDC-BRENT`
6	An equity price and news	`/v/listed-company/CO-AAPL`; `/v/news-topic/TOPIC-FED`
7	Compare two dates	on a trade, pick one of your dates → **Compare** → the other
8	BI by source system	**BI → Catalogue → Trades overview**; change a slicer
9	VaR and exposure on your data	`VAR BANK`, `EXP BANK` (section 8)
10	A history	`TRD <your id> 60` (two or more of your files), `TRD 1000279 60` (amended ten times); **C** changes only, **D** the fields that changed



1. An ENDUR deal



2. Forms → Trading

The pictures are of the synthetic data, loaded exactly as in 11.2.

Optional: the S3 tests, against simulators that need no `.exe` and no Docker: `pip install "moto[server]==5.2.3"`, then `.\mvnw.cmd -q -pl :drishti-identity,plugins/drishti-plugin-iceberg -am test "-Dtest=S3ContentStoreTest,S3MockContentStoreTest,MinioContentStoreTest,NativeOnS3SimulatorTest" "-Dsurefire.failIfNoSpecifiedTests=false"` and `& $py -m unittest -v tools\lake\test_s3_simulators.py`. The MinIO tests are skipped on Windows (they run a server binary).

Afterwards stop both, and check that `git status` (in a clone) shows nothing new outside `data\` and that your files never left `<private>`. Temporary files (`data\tmp`), caches (`data\cache`) and the grouped pages (`data\derived`) may be deleted while both are stopped.

12. Troubleshooting

What you see	Do this
`Drishti needs Java 21 or newer` (build or start)	`JAVA_HOME` (or `DRISHTI_JAVA_HOME`, or `--java-home`) at a JDK 21
`start-server.ps1 … cannot be loaded because running scripts is disabled`	run the `.cmd` (`bin\start-server.cmd`), or `powershell -ExecutionPolicy Bypass -File …`
PowerShell says `lake` is not recognized	`lake …` is a sub-command: `& $py "$tools\drishti.py" lake …`
`py -3.13 -m venv` fails on Linux with *ensurepip is not available*	`sudo apt install python3.13-venv`, or use `uv venv`
`Port 18480 (or 17480) is already in use`	another copy is running: `bin/status.sh` shows it; stop it (`bin/stop-server.sh`, `systemctl stop`, `launchctl unload`); more in 6.6
The console shows *Server not reachable*	the server is not up yet, or `--server` names the wrong address; `curl http://<host>:18480/actuator/health`
Every page empty; the console's log says `DRS-5010 missing bearer token`	the console and the server need the same `DRISHTI_TOKEN_SECRET`, and `DRISHTI_AUTH_ENABLED=true` on the console
The server refuses to start: `drishti.security.secret must be at least 32 bytes`	a longer `DRISHTI_TOKEN_SECRET`
Pages show the packs' samples, not the lake; quotes or news try the internet	`DRISHTI_DEMO_ENABLED=false` and `DRISHTI_OFFLINE=true` (or `--demo`)
`VAR BANK` says no data for today	the demo lake ends on another date: rebuild it (`lake demo`), or pick its last date in the date box
BI shows 0 trades while Admin → Lakes has the table	stop the server, delete `config\connectors` (the server writes it again from the packs), start it
`OutOfMemoryError`	a larger `-Xmx` (`--heap 8g`): about 4 GB per 400,000 trades a day
A page says `DRS-1004 … timed out` the first time	the first read of a date on a large table; open it again; exclude `data` from real-time antivirus scanning
documents are not cached on disk in the log	a second server uses the same data folder: one data folder per server
A module's pages are missing	`GET /api/v1/modules`: the module under `off` with its reason; switch it on and restart
A lens is not offered	`DRISHTI_LENSES` names it, and its modules are on

Everything else, with the error codes: OPERATIONS_GUIDE.md › Troubleshooting.

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